

Currency Mismatch under Earnings-Based Borrowing*

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Abstract

The conventional measure of corporate currency mismatch (FX) compares foreign-currency liabilities with foreign-currency assets. We show that this measure is incomplete when firms borrow against earnings following an exchange rate depreciation. In a two-sector small open economy model with dominant-currency pricing, an exchange rate depreciation tightens the non-tradable producer's borrowing constraint even at a zero dollar-debt share. A one-percent depreciation reduces non-tradable investment by about 0.33 percent even without dollar debt; at the average dollar share of non-tradable debt observed in the data for emerging markets, the contraction is about one-quarter larger. We derive a closed-form earnings-adjusted FX exposure index showing that the relevant measure of mismatch is net dollar operating exposure relative to pledgeable earnings. Because exchange rates and pledgeable earnings respond jointly to shocks, depreciations of the same size can have opposite effects on borrowing capacity depending on their source—a foreign financial tightening tightens borrowing constraints, whereas a fall in export demand can relax them—which may help explain why empirical studies that pool depreciation episodes find mixed evidence on whether depreciations are contractionary.

Keywords: FX mismatch, earnings-based borrowing constraints, dominant currency pricing, financial accelerator, Emerging markets

JEL Classification: F31, F38, E32, E44, G30.

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1 Introduction

A large literature asks whether foreign-currency liabilities make exchange-rate depreciations contractionary in emerging markets. Standard balance-sheet models, in which a depreciation raises the domestic value of dollar debt relative to firms' assets or collateral, predict that they can be (Céspedes et al., 2004). The empirical evidence, however, is mixed.¹ This paper asks whether the way firms borrow can help account for this ambiguity, and in particular why depreciations generated by different shocks can have different financial consequences. We argue that the usual formulation of the question misses an important dimension of corporate currency exposure: firms often borrow against *earnings*, and the exchange rate changes not only the domestic value of foreign-currency liabilities but also the operating cash flows against which firms borrow. As a result, FX mismatch is not generally a balance-sheet concept alone. Under earnings-based financing, it is also a cash-flow concept. This distinction matters because the currency composition of operating cash flows differs sharply across firms. Exporters often receive dollar revenues, while firms producing for the domestic market may purchase dollar-invoiced imported inputs but earn revenues in local currency. A depreciation therefore raises the domestic-currency value of export revenues but also the cost of imported inputs. If borrowing capacity is tied to EBITDA, these operating effects become financial effects: a depreciation can expand the borrowing capacity of a firm with sufficiently large dollar revenues, while reducing the borrowing capacity of a firm exposed to dollar input costs. Most strikingly, the latter effect arises even when the firm has *no foreign-currency debt*.

Two empirical facts motivate this mechanism. First, earnings-based covenants are pervasive in syndicated lending to emerging-market firms. In our Dealscan sample of covenanted EM syndicated loans, debt-to-EBITDA restrictions are the most prevalent covenant type, followed by interest-coverage covenants (see also Camara & Sangiaco, 2022; Drechsel, 2023; Lian & Ma, 2020).² The data suggest that, on average, firms can borrow up to about four times their annual earnings (3.89 times EBITDA, weighting by loan value), but sudden shocks to earnings can trigger covenant breaches, with consequences for firms' borrowing capacity, liquidity, and future earnings (Chava & Roberts, 2008; Roberts & Sufi, 2009). Second, under dominant-currency pricing, a large share of emerging-market exports and imported inputs is invoiced in dollars (Gopinath et al., 2020). Dollar export revenues can therefore provide a natural hedge against dollar liabil-

¹Aguiar (2005) finds that Mexican firms with greater exposure to short-term foreign-currency debt reduced investment more sharply following the 1994 devaluation. Kim et al. (2015) show that, during the Korean crisis, small firms with more short-term foreign-currency debt were more likely to exit and, conditional on survival, experienced larger declines in sales. By contrast, Bleakley and Cowan (2008) find no negative investment effect of dollar debt following depreciations in Latin America and show that firms tend to match the currency denomination of their liabilities to the exchange-rate sensitivity of their profits. Galindo et al. (2003) survey the earlier firm-level evidence.

²Firms too small to access syndicated loans often rely instead on credit from their suppliers. In emerging markets, large firms that borrow in foreign currency extend such trade credit to their domestic customers, so dollar funding conditions propagate along supply chains (Hardy & Saffie, 2024; Hardy et al., 2026), and the credit a supplier extends depends on the financial condition of both parties (Petersen & Rajan, 1997).

ities (Casas et al., 2023). With earnings-based borrowing, however, this hedge extends beyond debt revaluation: after a depreciation, dollar revenues also raise the earnings that support borrowing capacity, whereas dollar input costs reduce pledgeable earnings and create a financing exposure even in the absence of dollar debt. An appreciation reverses both effects.

We formalize this mechanism in a two-sector small open economy New Keynesian model. Firms own capital, finance themselves with debt and equity subject to quadratic dividend-adjustment costs as in Jermann and Quadrini (2012), and borrow subject to an earnings-based constraint calibrated to the debt-to-EBITDA covenants observed in Dealscan. Export prices are sticky in dollars, firms use imported intermediate inputs, and a fraction ϕ_b of debt is dollar-denominated. Tradable firms sell both at home and abroad, whereas non-tradable firms sell only domestically. The direct financial effect of an exchange-rate movement is summarized by a simple exposure statistic. Let

$$\chi = \frac{p^X y^X - m}{\Pi},$$

where $p^X y^X$ denotes dollar export revenues, m dollar-invoiced imported-input expenditure, and Π EBITDA. The numerator is the firm’s net dollar operating flow, so χ measures net dollar operating exposure per unit of pledgeable earnings. Because every firm also pays its workers in local currency, no firm’s operating cash flow is denominated in a single currency: even a multinational subsidiary that exports and imports entirely in dollars must pay local staff. χ captures this mix. Holding quantities and prices fixed, it is the elasticity of EBITDA with respect to the exchange rate, and local-currency costs enter through its denominator. For a firm whose revenues and inputs are both in dollars, local wages make EBITDA smaller than the net dollar flow, so $\chi > 1$: a depreciation raises its pledgeable earnings more than proportionally, and no dollar-debt share can offset the gain. Combining χ with the dollar share of debt ϕ_b gives our *earnings-adjusted FX exposure index*, $\mathcal{X} = \chi - \phi_b$ (Proposition 1): holding quantities and prices fixed, a depreciation raises the firm’s net financing headroom if and only if $\mathcal{X} > 0$. The index is a mechanical statistic built from model primitives rather than an equilibrium cutoff, but it closely predicts the sign of the equilibrium response in the quantitative model (Section 4.2). Hence the relevant measure of currency mismatch is not the dollar share of liabilities alone, but the joint currency composition of liabilities and operating cash flows.

This result has three implications. First, *earnings-based borrowing converts operating FX exposure into financing exposure*. A non-exporting firm that purchases dollar-invoiced inputs experiences higher production costs after a depreciation in any model. With an earnings-based constraint, however, the same increase in input costs also lowers EBITDA and therefore reduces the amount the firm can borrow. The borrowing constraint tightens even when the firm’s foreign-currency debt is zero. Conventional measures based on foreign-currency liabilities net of foreign-currency assets therefore miss an economically relevant source of financial FX exposure. Second, the effect depends on what is pledged. Under an asset-based borrowing constraint, the exchange rate revalues foreign-currency liabilities relative to collateral, but does not directly revalue the pledged collateral itself.

Under an earnings-based constraint, by contrast, the exchange rate changes the value of the pledged object. Dollar export revenues can therefore expand borrowing capacity after a depreciation, whereas dollar input costs reduce it. The distinction affects not only the magnitude but potentially the *sign* of the financial effect of a depreciation. Which sign obtains depends on a few ratios that can be measured in firm data. For an exporter that invoices in dollars, χ rises with export intensity and falls with the imported-input share, and a thin EBITDA margin magnifies it, because χ scales the net dollar flow by earnings. A depreciation directly improves the exporter’s financing position if and only if its dollar-debt share lies below χ . For a non-exporter, χ is negative, so no debt composition reverses the sign (Section 3.6). Third, a depreciation by itself is not sufficient to identify financial stress, because the shocks that move the exchange rate also move the earnings that firms pledge. What matters is the joint response of the exchange rate, pledgeable earnings, and foreign-currency debt. A foreign financial tightening, for example, weakens the currency while lowering non-tradable earnings; a fall in export demand also weakens the currency but lowers domestic wages and thereby raises those earnings. The same exchange-rate movement can therefore have different financial consequences depending on the shock that generates it. This provides a potential explanation for the mixed empirical evidence on contractionary depreciations: pooling depreciation episodes generated by different shocks can combine observations in which currency mismatch strongly amplifies financial stress with others in which the revaluation of dollar liabilities is offset by higher earnings, or even reversed.

We quantify these forces in a two-sector economy in which both tradable and non-tradable firms borrow against earnings and use imported intermediate inputs, but only tradable firms earn dollar revenues. Following a foreign interest-rate tightening, the earnings-based constraint amplifies the shock through firms’ borrowing and investment rather than through the exchange rate: the depreciation and the policy response are almost identical whatever the currency composition of firms’ debt, while borrowing conditions and investment diverge sharply (Section 4.1). The depreciation tightens the non-tradable firm’s borrowing constraint even when its dollar-debt share is zero, because the higher domestic-currency cost of imported inputs lowers pledgeable earnings. By contrast, sufficiently large dollar export revenues protect the tradable firm’s borrowing capacity. The covenant itself offers a direct way to read how much a depreciation tightens borrowing. Holding quantities fixed, a firm’s debt-to-EBITDA ratio rises by $-\mathcal{X}$ percent for each percent of depreciation. A 10 percent depreciation therefore pushes a non-tradable firm that starts at the 3.89 limit to about 4.03 times EBITDA at the observed dollar share, so that it must cut debt by about 3.4 percent to remain compliant, and to about 4.27 at the stress share of 75 percent, requiring a cut of about 9 percent. The same depreciation lowers an exporter’s ratio to about 3.70, leaving it room to borrow about 5 percent more. The quantitative effects are economically meaningful. At the sectoral dollar-debt shares observed in the data, the investment contraction in the non-tradable sector is about one quarter larger than in a counterfactual without dollar debt. At a stress value of 75 percent for the dollar share of non-tradable debt the contraction more than doubles, whereas tradable investment changes little. These differences arise even though

both sectors face the same exchange rate, interest rates, aggregate shocks, and ex ante borrowing-constraint tightness. What differs is the currency composition of their operating cash flows and liabilities, and the experiments separate the two: the zero-dollar-debt counterfactual isolates the cash-flow channel, and raising the non-tradable dollar share then adds the liability channel on top of an essentially unchanged operating shock.

We next show that the financial consequences of an exchange-rate movement depend on the underlying shock. A foreign financial tightening and an export-demand contraction both generate a depreciation, but they produce opposite movements in the non-tradable borrowing constraint per unit of depreciation. Following the financial tightening, earnings weaken while dollar liabilities appreciate in domestic-currency terms, tightening the constraint. Following the export-demand shock, lower inflation induces monetary easing and lowers domestic production costs sufficiently to raise non-tradable earnings relative to debt revaluation, loosening the constraint. An import-cost shock provides a complementary case: although it contracts activity strongly through higher production costs, the associated inflation induces a monetary tightening and an exchange-rate appreciation, which reduces the domestic value of dollar liabilities and relaxes the financial constraint on impact. The case for leaning against a depreciation to protect balance sheets may therefore also depend on what caused it.

The general lesson therefore does not depend on the label attached to a shock but on what the shock does to earnings: *FX mismatch amplifies exchange-rate movements that coincide with weak pledgeable earnings*. The model nevertheless delivers a falsifiable empirical prediction. Financial amplification from currency mismatch should be particularly strong after currency depreciations driven by financial shocks—such as those associated with global financial tightening, capital-flow reversals, or foreign monetary policy shocks—and weaker, absent, or potentially reversed following exchange-rate movements generated by shocks that improve domestic operating margins. This mechanism also creates a monetary-policy trade-off. Stabilizing the exchange rate protects firms whose borrowing capacity deteriorates after a depreciation, particularly non-tradable firms exposed to dollar input costs. At the same time, the interest-rate increase needed to stabilize the currency depresses aggregate demand. An exchange-rate-augmented Taylor rule therefore dampens the deterioration in non-tradable balance sheets but deepens the aggregate contraction.

Related literature. Our paper connects four strands of research. First, the literature on earnings-based borrowing constraints shows that borrowing against cash flows rather than collateral changes macroeconomic transmission and policy trade-offs (Drechsel, 2023; Drechsel & Kim, 2024; Greenwald, 2019; Lian & Ma, 2020). In an emerging-market setting, Camara and Sangiacomo (2022) document cash-flow-based lending and show that such constraints amplify foreign interest-rate shocks. We show that, once pledgeable earnings contain cash flows denominated in different currencies, exchange-rate movements can change the *sign* as well as the magnitude of financial amplification.

Second, we relate to the corporate currency-mismatch literature. Casas et al. (2023) show that dollar export revenues hedge firms against the revaluation of dollar liabili-

ties, with the financing effects of depreciations concentrated among non-exporters. Our mechanism operates through an additional margin: when firms borrow against earnings, the exchange rate revalues the object that supports borrowing capacity. Dollar revenues therefore hedge not only debt repayment but also borrowing capacity, while dollar input costs generate financial exposure even in firms without foreign-currency liabilities. Using Korean firm-level data, Kim and Lee (2026) show that the balance-sheet effect of foreign-currency debt shapes exchange-rate pass-through, and Kim et al. (2026) examine whether exporters’ dollar revenues in fact hedge their dollar debt. Dollar funding conditions also propagate through firm-to-firm credit, as large firms that borrow in foreign currency extend trade credit to domestic customers (Hardy & Saffie, 2024; Hardy et al., 2026).

Third, we connect the dominant-currency paradigm (Amiti et al., 2022; Gopinath et al., 2020) to corporate finance. That literature emphasizes how invoicing currency shapes price adjustment, pass-through, and expenditure switching. In our framework, invoicing currency also becomes a financing variable because it determines how exchange-rate movements affect pledgeable operating earnings.

Finally, we contribute to the classic emerging-market balance-sheet literature, which measures currency mismatch primarily through the currency composition of assets and liabilities. Our results imply that this stock-based measure is incomplete when borrowing depends on cash flow. The relevant object also includes the currency composition of revenues and variable costs. This broader measure helps determine not only which firms are exposed to depreciations, but also when exchange-rate movements translate into financial stress. The small-open-economy closure follows Schmitt-Grohé and Uribe (2003), while our trade-shock exercises relate to the fragmentation analysis of Ambrosino et al. (2026).

2 Motivating evidence

We briefly document three facts; sample construction and additional figures are in Appendix M.

Fact 1: earnings-based covenants are widespread. In Dealscan syndicated loans to EM borrowers featuring covenants, the most prevalent covenant type caps total debt at a multiple of EBITDA, present in almost 60 percent of covenanted loan value; its loan-value-weighted mean limit is 3.89 (median 3.50). Interest-coverage covenants are the second most common. We use the mean 3.89 to calibrate the tightness of the EBC, matching the *face value* of debt to annual EBITDA exactly, and note that our sample establishes prevalence among covenanted syndicated loans, not an economy-wide shift from asset-based to earnings-based borrowing.

Fact 2: dollar debt is pervasive and unevenly held. Capital IQ instrument-level debt data for 10,292 public non-financial firms across 29 EM countries show a substantial dollar share in corporate debt (Figure 1). Crucially for our mechanism, dollar borrowing is not confined to exporters: Table 2 splits the currency composition by tradability, classifying industries by their export orientation in OECD TiVA data. Non-tradable firms

Table 1: Most pervasive loan covenant types, values and frequency

Covenant type	p25	Median	p75	Mean	Frequency
Max. Debt to EBITDA	3.00	3.50	4.50	3.89	57.9%
Min. Interest Coverage	2.00	2.50	3.00	2.86	51.7%
Min. Debt Service Coverage	1.20	1.25	1.55	1.56	13.4%
Tangible Net Worth	425M	5B	16B	54.6B	13.1%
Max. Leverage ratio	0.60	1.10	3.00	2.31	12.1%
Max. Debt to Tangible Net Worth	1.20	1.50	2.10	2.39	10.3%

Note: Authors’ calculations from Thomson Reuters LPC Dealscan following Drechsel (2023). Covenants with frequency above 10% are included. Sample: loan deals with at least one covenant issued 1993–2020 by EM nonfinancial corporations (373 deals, 262 distinct borrowers). Mean and frequency weighted by real loan size; ‘M’ and ‘B’ are millions and billions of 2009 US dollars.

Table 2: EM firm debt currency composition by tradability

Industry	USD	Local currency	Other currency	Euro
Non-tradable	15.75	65.28	15.87	3.08
Tradable	28.21	52.53	15.24	4.01

Note: Instrument-level debt of 10,292 public non-financial firms, 29 EM countries by headquarters, 2002–2022. Authors’ calculations from Capital IQ; cleaning follows Kim (2019). Tradable industries are mining, agriculture, manufacturing and transport, classified by export orientation $X_{c,s} = \text{Exports}_{c,s} / \text{GrossOutput}_{c,s}$ in OECD TiVA (ICIO) data, 1995–2020; non-tradable is every other non-financial industry.

hold about 16 percent of their debt in dollars while earning local-currency revenues — since they pay for dollar-invoiced inputs but earn no dollar revenue, the model’s exposure index (Section 3.6) is negative for them whatever the composition of their debt — whereas tradable firms’ 28 percent dollar share sits well below their net dollar operating exposure in the model, $\chi_T \approx 0.78$, the dollar-debt share at which the hedge provided by their export revenue would be exhausted. The evidence is consistent with the mechanism binding where the model says it does — on the non-tradable sector — though two caveats apply: the TiVA-based classification is coarse (some services firms classified as non-tradable do export, so “no dollar revenue” is a model assumption rather than a measured fact), and debt denomination is not net FX exposure if firms hedge with derivatives; evidence of limited operational matching (Alfaro et al., 2023) does not by itself rule out financial hedging.

3 Model

We consider a two-sector small open economy with tradable (T) and non-tradable (N) production. Firms in both sectors use capital, labor, and imported intermediate inputs,

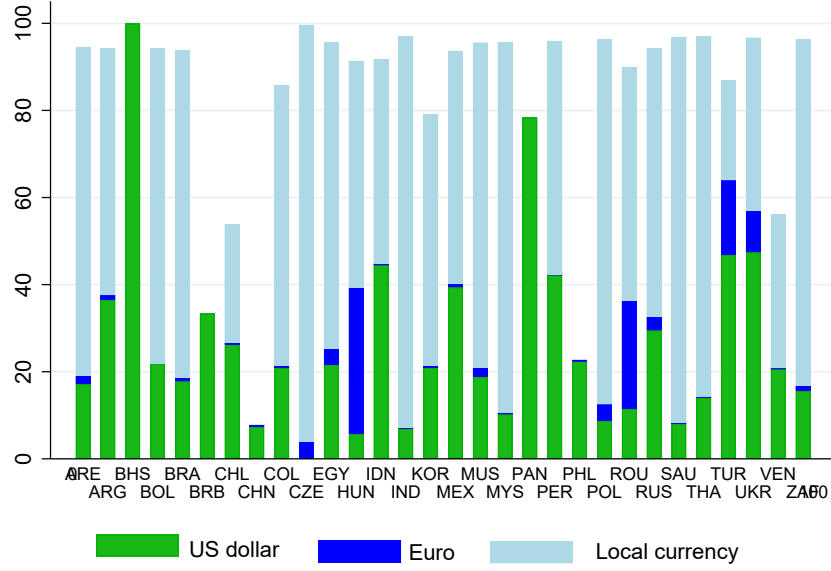


Figure 1: Currency denomination of EM corporate debt. Instrument-level debt of 10,292 public non-financial firms, 29 EM countries by headquarters. Authors’ calculations from Capital IQ; cleaning follows Kim (2019). The residual “other currency” category of Table 2 is omitted from the figure.

accumulate capital, and borrow in domestic and foreign currency subject to an earnings-based borrowing constraint. Tradable firms sell both domestically and abroad, whereas non-tradable firms sell only in the domestic market. The difference in the currency composition of revenues generates heterogeneous exposure to exchange-rate movements. We present firms first, since the earnings-based borrowing constraint is the model’s one non-standard ingredient; the household block, monetary policy, and market clearing are standard. Time is quarterly. All domestic prices are expressed relative to the domestic CPI. An increase in the real exchange rate s_t denotes a real depreciation.

Notation. We follow international-macro conventions throughout: σ is risk aversion and φ the inverse Frisch elasticity; η the home–foreign substitution elasticity within tradables and ϵ the elasticity across varieties, as in Galí and Monacelli, 2005; ω_N and η_N the non-tradable expenditure weight and the tradable–non-tradable elasticity; σ_x the aggregate trade elasticity (Gopinath et al., 2020); $mc_{j,t}$ sector j ’s real marginal cost; κ_I the investment adjustment cost and κ_p the Rotemberg price-adjustment cost; θ_π and θ_k the earnings- and asset-based pledgeability parameters (Drechsel, 2023); ψ_d the dividend adjustment cost (Jermann & Quadrini, 2012); κ_D the debt-elastic-premium elasticity (Schmitt-Grohé & Uribe, 2003); and ϕ_b^j sector j ’s dollar share of debt. r_t^H, r_t^F are net nominal rates on one-period bonds, Ψ_t the borrowing premium, $\Phi(d_{j,t})$ the dividend adjustment-cost function, and $\gamma_{j,t} \equiv 1 + \Phi'(d_{j,t})$.

3.1 Firms

We begin with firms, because the earnings-based borrowing constraint is the model's key ingredient. The production side consists of competitive final-good aggregators and monopolistically competitive intermediate producers in the tradable and non-tradable sectors. Intermediate producers finance themselves with debt and equity and can borrow only up to a multiple of their current earnings, $B_{j,t} \leq \theta_\pi \Pi_{j,t}$ (equation (15) below). Because part of those earnings and part of their debt are denominated in dollars, the exchange rate moves both sides of this constraint. Firms are owned by households and discount future payouts with the household stochastic discount factor $\Lambda_{t,t+1}$; CPI inflation π_t and the country premium Ψ_t are defined in Section 3.2.

3.1.1 Final good producers

For $\ell \in \{H, X, N\}$, final-good producers aggregate differentiated varieties according to

$$y_t^\ell = \left[\int_0^1 (y_{i,t}^\ell)^{\frac{\epsilon-1}{\epsilon}} di \right]^{\frac{\epsilon}{\epsilon-1}}, \quad y_{i,t}^\ell = \left(\frac{p_{i,t}^\ell}{p_t^\ell} \right)^{-\epsilon} y_t^\ell. \quad (1)$$

Aggregate foreign demand is

$$y_t^X = \gamma_z y_t^* (q_t^X)^{-\sigma_x}, \quad q_t^X = \begin{cases} p_t^X, & \text{DCP,} \\ p_t^H / s_t, & \text{PCP.} \end{cases} \quad (2)$$

Under dominant currency pricing (DCP), the empirically prevalent convention for emerging-market exports (Gopinath et al., 2020), depreciation raises the domestic value of export revenues without directly changing the foreign-currency price faced by importers, so export *quantities* are insulated by the sticky dollar price and the exchange rate transmits through revenue revaluation. Under producer currency pricing (PCP), the classic benchmark of the open-economy literature (Galí & Monacelli, 2005), the firm sets a sticky price in domestic currency, so a depreciation lowers the foreign-currency price p_t^H / s_t one-for-one and stimulates export demand through expenditure switching with elasticity σ_x ; the local-currency value of a given sale is unchanged, so the exchange rate transmits through quantities rather than revaluation.

3.1.2 Intermediate producers

Intermediate firms in both sectors produce output and use the same production technology by employing capital, labor, and foreign imports,

$$y_{j,i,t} = z_{j,t} k_{j,i,t-1}^\alpha N_{j,i,t}^{1-\alpha-\alpha_m} m_{j,i,t}^{\alpha_m}, \quad j \in \{T, N\}, \quad (3)$$

with the only difference that tradable intermediate producers sell their output both domestically and abroad:

$$y_{T,i,t} = y_{i,t}^H + y_{i,t}^X. \quad (4)$$

Firms take factor prices as given, so conditional cost minimization equates each factor price to its marginal product valued at marginal cost:

$$w_t = mc_{j,i,t}(1 - \alpha - \alpha_m) \frac{y_{j,i,t}}{N_{j,i,t}}, \quad (5)$$

$$s_t = mc_{j,i,t} \alpha_m \frac{y_{j,i,t}}{m_{j,i,t}}. \quad (6)$$

Here $mc_{j,i,t}$ is the multiplier on the production constraint, that is, sector j 's real marginal cost. Under DCP, operating earnings are

$$\Pi_{T,i,t} = p_{i,t}^H y_{i,t}^H + s_t p_{i,t}^X y_{i,t}^X - w_t N_{T,i,t} - s_t m_{T,i,t}, \quad (7)$$

$$\Pi_{N,i,t} = p_{i,t}^N y_{i,t}^N - w_t N_{N,i,t} - s_t m_{N,i,t}. \quad (8)$$

These are operating earnings before interest, taxes, and depreciation. This is the object the covenant pledges, so its currency composition determines borrowing capacity. The two sectors differ in exactly that composition. The tradable firm's earnings carry dollars on *both* sides, revenue $s_t p_{i,t}^X y_{i,t}^X$ and costs $s_t m_{T,i,t}$, so a depreciation moves Π_T by the *net* dollar operating flow $p^X y^X - m_T$, which may be positive. The non-tradable firm sells at a local-currency price and buys dollar inputs, so dollars enter its earnings only as a cost and a depreciation lowers Π_N by m_N unambiguously.

Price adjustment Rotemberg, 1982 Under DCP, tradable firms incur

$$AC_{T,i,t}^p = \frac{\kappa_p}{2} (\pi_{i,t}^H - 1)^2 p_t^H y_t^H + \frac{\kappa_p}{2} (\pi_{i,t}^X - 1)^2 s_t p_t^X y_t^X, \quad (9)$$

where

$$\pi_{i,t}^H = \frac{p_{i,t}^H}{p_{i,t-1}^H} \pi_t, \quad \pi_{i,t}^X = \frac{p_{i,t}^X}{p_{i,t-1}^X} \pi_t^*.$$

Non-tradable firms incur

$$AC_{N,i,t}^p = \frac{\kappa_p}{2} (\pi_{i,t}^N - 1)^2 p_t^N y_t^N, \quad \pi_{i,t}^N = \frac{p_{i,t}^N}{p_{i,t-1}^N} \pi_t. \quad (10)$$

Both adjustment costs are *resource* costs paid in units of the final consumption basket — the sectoral revenue terms $p_t^H y_t^H$, $s_t p_t^X y_t^X$, and $p_t^N y_t^N$ only scale their magnitude. They therefore enter aggregate absorption in (42) alongside consumption and investment, and are allocated across goods by the same CES expenditure system. The corresponding pricing conditions are reported in Appendix B.

Financing and borrowing constraints. Henceforth we impose symmetric equilibrium within each sector and suppress the firm index i . Each sector has its own dividends, investment, capital stock, and debt. Dividends are subject to quadratic adjustment costs following Jermann and Quadrini, 2012

$$\Phi(d_{j,t}) = \psi_d (d_{j,t} - \bar{d}_j)^2. \quad (11)$$

Total face-value debt is

$$B_{j,t} = b_{j,t}^H + s_t b_{j,t}^F, \quad (12)$$

with sector-specific foreign-currency share

$$b_{j,t}^H = (1 - \phi_b^j) B_{j,t}, \quad s_t b_{j,t}^F = \phi_b^j B_{j,t}. \quad (13)$$

Because debt is one-period, the entire stock is reissued each period and ϕ_b^j is a share of *new issuance*; the currency composition of the inherited stock is predetermined within the period, and it is this inherited composition that a depreciation revalues. Fixing the share avoids the portfolio indeterminacy that endogenous currency choice would create at first order. The sector flow-of-funds constraint is

$$d_{j,t} + \Phi(d_{j,t}) + i_{j,t} + AC_{j,t}^p + \frac{b_{j,t-1}^H}{\pi_t} + \frac{s_t b_{j,t-1}^F}{\pi_t^*} = \Pi_{j,t} + \frac{b_{j,t}^H}{1 + r_t^H(1 - \tau)} + \frac{s_t b_{j,t}^F}{1 + [(1 + r_t^F)\Psi_t - 1](1 - \tau)}. \quad (14)$$

Debt carries a tax advantage (Jermann and Quadrini, 2012): interest is deductible at rate τ . The tax treatment is identical across currencies, so the currency composition of debt is not driven by taxes; the government finances the subsidy with lump-sum taxes. The tax wedge is also what makes the borrowing constraint bind: debt is strictly cheaper than internal funds. In the baseline earnings-based constraint (EBC) economy,

$$B_{j,t} \leq \theta_\pi \Pi_{j,t}, \quad j \in \{T, N\}. \quad (15)$$

Borrowing capacity is a multiple θ_π of the sector's current operating earnings — the debt-to-EBITDA covenant of Table 1 — so it moves one-for-one with pledged cash flow, and hence with the exchange rate through (7)–(8). The two sectoral constraints are distinct and may therefore bind in different states. For comparison, under the asset-based constraint (ABC),

$$B_{j,t} \leq \theta_k(1 - \delta)\mathbb{E}_t[q_{j,t+1}]k_{j,t}. \quad (16)$$

Here capacity is instead a fraction θ_k of the depreciated resale value of installed capital, as in Kiyotaki and Moore (1997), so it tracks asset values rather than earnings; θ_k is chosen so that steady-state debt matches the EBC economy's, leaving the two economies to differ only in *what is pledged*.

Capital accumulation and firm optimality. Capital evolves according to

$$k_{j,t} = (1 - \delta)k_{j,t-1} + \left[1 - \frac{\kappa_I}{2} \left(\frac{i_{j,t}}{i_{j,t-1}} - 1\right)^2\right] i_{j,t}. \quad (17)$$

Adjustment costs are paid on the *growth* rate of investment, as in Smets and Wouters (2007).

Define

$$\gamma_{j,t} = 1 + \Phi'(d_{j,t}), \quad M_{j,t,t+1}^f = \Lambda_{t,t+1} \frac{\gamma_{j,t}}{\gamma_{j,t+1}}, \quad (18)$$

The first, $\gamma_{j,t}$, is the marginal cost of distributing a unit of earnings as dividends; its inverse $1/\gamma_{j,t}$ is the shadow value of a unit of internal funds, which rises above one precisely when the firm is forced to cut payouts below target. The second, $M_{j,t,t+1}^f$, is the firm's effective stochastic discount factor: it coincides with the household's kernel $\Lambda_{t,t+1}$ when internal funds are equally valuable in both periods, and departs from it whenever they are not — a firm whose funds are scarce today relative to tomorrow discounts the future more heavily than its owners do. Finally, $\zeta_{j,t}$ measures the value of relaxing the borrowing limit by one unit, expressed per unit of internal funds; it is zero whenever the constraint is slack.

Under the EBC, the capital FOC is

$$q_{j,t} = \mathbb{E}_t \left[M_{j,t,t+1}^f \left\{ (1 + \theta_\pi \zeta_{j,t+1}) mc_{j,t+1} \alpha \frac{y_{j,t+1}}{k_{j,t}} + (1 - \delta) q_{j,t+1} \right\} \right]. \quad (19)$$

This is a standard q -theory relation — the value of installed capital equals the discounted sum of its marginal product and its undepreciated resale value — with one addition. The marginal product is multiplied by $1 + \theta_\pi \zeta_{j,t+1}$ rather than by one, because a unit of earnings generated next period is worth its cash value *and* relaxes next period's constraint by θ_π units, each valued at $\zeta_{j,t+1}$. Capital therefore earns a financial premium on top of its physical return whenever the constraint is expected to bind, and the premium is forward-looking: what matters is the tightness of *future* constraints, since capital installed today produces pledgeable earnings from tomorrow onward. Under the ABC this term is absent — earnings are not pledged — and capital instead relaxes the constraint directly through its collateral value. The debt FOC is

$$\begin{aligned} \zeta_{j,t} = (1 - \phi_b^j) & \left[\frac{1}{1 + r_t^H (1 - \tau)} - \mathbb{E}_t \left(M_{j,t,t+1}^f \frac{1}{\pi_{t+1}} \right) \right] \\ & + \phi_b^j \left[\frac{1}{1 + [(1 + r_t^F) \Psi_t - 1] (1 - \tau)} - \mathbb{E}_t \left(M_{j,t,t+1}^f \frac{s_{t+1}}{s_t \pi_{t+1}^*} \right) \right]. \end{aligned} \quad (20)$$

Each bracket is the *financing wedge* on one currency: the proceeds raised today per unit of face value promised, minus the expected discounted cost of repaying it. A positive wedge means the firm would issue more debt if it could, so the constraint binds and $\zeta_{j,t} > 0$ measures by how much; the multiplier is the share-weighted average of the two wedges, since the firm issues in fixed proportions $1 - \phi_b^j$ and ϕ_b^j . The home-currency wedge is eroded by inflation, which lowers the real repayment $1/\pi_{t+1}$. The dollar wedge instead contains the expected real depreciation s_{t+1}/s_t : an exchange rate expected to depreciate raises the cost of servicing new dollar debt and compresses the wedge. This is the *marginal* exchange-rate channel and should be distinguished from the revaluation of *inherited* dollar liabilities in (14), which depends on the level of s_t rather than its expected path: the first operates on new issuance, the second drains internal funds directly.

3.2 Households

Household consumption is a CES aggregate of tradable and non-tradable goods,

$$C_t = \left[(1 - \omega_N)^{1/\eta_N} C_{T,t}^{\frac{\eta_N-1}{\eta_N}} + \omega_N^{1/\eta_N} C_{N,t}^{\frac{\eta_N-1}{\eta_N}} \right]^{\frac{\eta_N}{\eta_N-1}}, \quad (21)$$

where ω_N is the non-tradable expenditure weight and η_N is the elasticity of substitution between tradable and non-tradable goods. The tradable composite is

$$C_{T,t} = \left[(1 - \omega)^{1/\eta} C_{H,t}^{\frac{\eta-1}{\eta}} + \omega^{1/\eta} C_{F,t}^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}}, \quad (22)$$

where $C_{H,t}$ and $C_{F,t}$ denote home-produced and imported tradable goods with their elasticity of substitution η . Let p_t^N , p_t^T , and p_t^H denote prices relative to the domestic CPI. The real price of the imported good is s_t . The price indices satisfy

$$1 = (1 - \omega_N)(p_t^T)^{1-\eta_N} + \omega_N(p_t^N)^{1-\eta_N}, \quad (23)$$

$$(p_t^T)^{1-\eta} = (1 - \omega)(p_t^H)^{1-\eta} + \omega s_t^{1-\eta}. \quad (24)$$

Cost minimization implies

$$C_{N,t} = \omega_N (p_t^N)^{-\eta_N} C_t, \quad C_{T,t} = (1 - \omega_N) (p_t^T)^{-\eta_N} C_t, \quad (25)$$

$$C_{H,t} = (1 - \omega) \left(\frac{p_t^H}{p_t^T} \right)^{-\eta} C_{T,t}, \quad C_{F,t} = \omega \left(\frac{s_t}{p_t^T} \right)^{-\eta} C_{T,t}. \quad (26)$$

Let P_t denote the domestic consumer price index — the price of the consumption basket (21), which serves as the numeraire throughout — and P_t^* the foreign price level. Gross CPI inflation and gross foreign inflation are $\pi_t \equiv P_t/P_{t-1}$ and $\pi_t^* \equiv P_t^*/P_{t-1}^*$; we normalize $\pi_t^* = 1$, so the foreign price level is constant. Letting $\mathcal{E}_t$ denote gross nominal depreciation, the real exchange rate satisfies

$$\frac{s_t}{s_{t-1}} = \mathcal{E}_t \frac{\pi_t^*}{\pi_t}. \quad (27)$$

The representative household chooses consumption, supplies labor at the real wage w_t and holds domestic and foreign bonds $\{C_t, N_t, a_t^H, a_t^F\}_{t=0}^\infty$ to maximize

$$\mathbb{E}_0 \sum_{t=0}^\infty \beta^t \left[\frac{C_t^{1-\sigma} - 1}{1-\sigma} - \kappa_L \frac{N_t^{1+\varphi}}{1+\varphi} \right]. \quad (28)$$

Using a face-value convention for bond holdings, the household budget constraint is

$$C_t + \frac{a_t^H}{1 + r_t^H} + \frac{s_t a_t^F}{(1 + r_t^F) \Psi_t} + \mathcal{T}_t = w_t N_t + \frac{a_{t-1}^H}{\pi_t} + \frac{s_t a_{t-1}^F}{\pi_t^*} + d_{T,t} + d_{N,t}, \quad (29)$$

where $\mathcal{T}_t$ is a lump-sum tax financing the corporate debt tax advantage, and $d_{T,t}, d_{N,t}$ denote the dividends that households get from holding the tradable and non-tradable firms. Net foreign assets, expressed in foreign-good units, correspond to households' foreign bond holdings and the corporate sector foreign financing:

$$\mathcal{B}_t = a_t^F - b_{T,t}^F - b_{N,t}^F, \quad (30)$$

International borrowing is subject to a debt-elastic country premium. With a constant discount factor and incomplete international financial markets, the small open economy's net foreign asset position would otherwise inherit a unit root, leaving no stationary steady state to approximate around. Following Schmitt-Grohé and Uribe (2003), we therefore make the effective cost of foreign funds increasing in the economy's net external indebtedness,

$$\Psi_t = \exp \left\{ -\kappa_D (\mathcal{B}_t - \bar{\mathcal{B}}) \right\} \xi_t, \quad (31)$$

where $\bar{\mathcal{B}}$ is the steady-state net foreign asset position, $\kappa_D > 0$ the semi-elasticity of the premium to it, and ξ_t an exogenous AR(1) country-risk shock.

Defining the household stochastic discount factor as

$$\Lambda_{t,t+1} = \beta \left(\frac{C_{t+1}}{C_t} \right)^{-\sigma}, \quad (32)$$

the household optimality conditions are

$$\kappa_L N_t^\varphi = C_t^{-\sigma} w_t, \quad (33)$$

$$1 = \mathbb{E}_t \left[\Lambda_{t,t+1} \frac{1 + r_t^H}{\pi_{t+1}} \right], \quad (34)$$

$$1 = \mathbb{E}_t \left[\Lambda_{t,t+1} \Psi_t (1 + r_t^F) \frac{s_{t+1}}{s_t \pi_{t+1}^*} \right]. \quad (35)$$

Combining (34) and (35) yields the uncovered interest parity (UIP) condition

$$\mathbb{E}_t \left[\Lambda_{t,t+1} \left(\frac{1 + r_t^H}{\pi_{t+1}} - \Psi_t (1 + r_t^F) \frac{s_{t+1}}{s_t \pi_{t+1}^*} \right) \right] = 0, \quad (36)$$

so that, up to first order, the expected real depreciation equals the interest differential adjusted for the country premium Ψ_t : a rise in the foreign rate or in the premium requires an expected appreciation along the transition, which the exchange rate delivers by depreciating on impact. Deviations from risk-free UIP are governed entirely by Ψ_t , which is the object the risk-premium experiments of Section 4 perturb.

3.3 Monetary policy

The monetary authority sets the domestic interest rate according to

$$r_t^H = \rho_R r_{t-1}^H + (1 - \rho_R) [\bar{r} + \phi_\pi (\pi_t - 1) + \phi_e (\mathcal{E}_t - 1)]. \quad (37)$$

The baseline sets $\phi_e = 0$; policy experiments allow $\phi_e > 0$. The foreign interest rate follows

$$r_t^F - \bar{r}^F = \rho_F(r_{t-1}^F - \bar{r}^F) + \varepsilon_t^F, \quad (38)$$

with foreign inflation normalized to one as above. The remaining exogenous states follow AR(1) processes in logs:

$$\log z_{j,t} = (1 - \rho_z) \log \bar{z} + \rho_z \log z_{j,t-1} + \varepsilon_{j,t}^z, \quad \log y_t^* = \rho_{y^*} \log y_{t-1}^* + \varepsilon_t^{y^*}, \quad \log \xi_t = \rho_{y^*} \log \xi_{t-1} + \varepsilon_t^\xi, \quad (39)$$

with $j \in \{T, N\}$; the foreign-demand and risk-premium processes share the persistence ρ_{y^*} , and $y^* = \xi = 1$ in steady state.

3.4 Trade-fragmentation shocks

We also perturb the cost of importing, for two reasons. Such a shock acts directly on the object at the centre of the mechanism — the dollar input bill $s_t m_{j,t}$ inside pledged earnings — yet, unlike a financial shock, it moves the exchange rate only through the monetary response to the inflation it creates. It therefore separates a shock that damages dollar-cost-exposed firms from one that damages them through FX revaluation, and it speaks directly to the policy debate on trade fragmentation. The main experiment is a delivered import-cost wedge — a fragmentation cost in the interpretation of Ambrosino et al. (2026), or an import tariff whose revenue is dissipated abroad:

$$s_t^m = s_t(1 + \tau_t^m), \quad \tau_t^m = \rho_\tau \tau_{t-1}^m + \varepsilon_t^m. \quad (40)$$

The delivered price s_t^m replaces s_t wherever imported goods are purchased: in final import demand, in the tradable price index (24), and in the imported-input bills of both sectors. For the combined fragmentation experiment of Appendix K we add a foreign export tariff, $\tau_t^x = \rho_\tau \tau_{t-1}^x + \varepsilon_t^x$, which under DCP changes foreign demand to

$$y_t^X = \gamma_z y_t^* [(1 + \tau_t^x) p_t^X]^{-\sigma_x}; \quad (41)$$

it is paid by the foreign buyer and never enters domestic accounts.

3.5 Equilibrium and market clearing

Because investment and adjustment costs are purchased in units of the same final-good basket, aggregate absorption is

$$A_t = C_t + i_{T,t} + i_{N,t} + \sum_{j \in \{T, N\}} [\Phi(d_{j,t}) + AC_{j,t}^p]. \quad (42)$$

Let $A_{N,t}$ and $A_{T,t}$ denote non-tradable and tradable absorption, with $A_{H,t}$ and $A_{F,t}$ the home and imported components of tradable absorption. The same CES expenditure

system implies

$$A_{N,t} = \omega_N (p_t^N)^{-\eta_N} A_t, \quad A_{T,t} = (1 - \omega_N) (p_t^T)^{-\eta_N} A_t, \quad (43)$$

$$A_{H,t} = (1 - \omega) \left(\frac{p_t^H}{p_t^T} \right)^{-\eta} A_{T,t}, \quad A_{F,t} = \omega \left(\frac{s_t}{p_t^T} \right)^{-\eta} A_{T,t}. \quad (44)$$

Goods markets clear according to

$$y_t^N = A_{N,t}, \quad y_t^H = A_{H,t}, \quad y_t^X = \gamma_z y_t^* (p_t^X)^{-\sigma_x}, \quad (45)$$

under DCP. Total imports are

$$\mathcal{M}_t = A_{F,t} + m_{T,t} + m_{N,t}. \quad (46)$$

Labor is mobile across sectors,

$$N_t = N_{T,t} + N_{N,t}, \quad (47)$$

and the domestic bond market clears according to

$$a_t^H = b_{T,t}^H + b_{N,t}^H. \quad (48)$$

GDP and the trade balance are

$$GDP_t = p_t^N y_t^N + p_t^H y_t^H + s_t p_t^X y_t^X - s_t (m_{T,t} + m_{N,t}), \quad (49)$$

$$TB_t = s_t p_t^X y_t^X - s_t \mathcal{M}_t. \quad (50)$$

The consolidated external budget constraint is

$$\frac{s_t \mathcal{B}_t}{(1 + r_t^F) \Psi_t} = \frac{s_t \mathcal{B}_{t-1}}{\pi_t^*} + TB_t. \quad (51)$$

These conditions imply

$$GDP_t = A_t + TB_t, \quad (52)$$

which is not imposed separately.

Equilibrium. Given the exogenous processes $\{z_{T,t}, z_{N,t}, y_t^*, r_t^F, \xi_t\}$, the monetary-policy rule, and initial conditions for capital, debt, investment, and prices, a competitive equilibrium is a collection of state-contingent sequences for prices, allocations, and multipliers such that (i) the household maximizes (28) subject to (29); (ii) firms in each sector maximize shareholder value subject to technology, demand, capital accumulation, the flow of funds, and the relevant borrowing constraint; (iii) final-good producers minimize expenditure; (iv) the monetary rule (37) holds and the government budget balances; and (v) the markets for the home and export tradable goods, the non-tradable good, labor, domestic bonds, and foreign bonds all clear. Appendices A–B list the full equation system.

3.6 FX Exposure under Earnings-Based Borrowing

Holding quantities fixed, a depreciation changes pledgeable earnings according to

$$\frac{\partial \Pi_{T,t}}{\partial \log s_t} = p_t^X y_t^X - m_{T,t}, \quad \frac{\partial \Pi_{N,t}}{\partial \log s_t} = -m_{N,t}.$$

It also raises the domestic value of inherited foreign-currency debt by $\phi_b^j B_j = \phi_b^j \theta_\pi \Pi_j$ when the EBC binds. Although the first effect changes the borrowing limit and the second the repayment burden, both affect the firm's net financing position and can be expressed relative to pledgeable earnings.

Proposition 1 (Earnings-adjusted FX exposure). *Define sector j 's net dollar operating exposure,*

$$\chi_T = \frac{p^X y^X - m_T}{\Pi_T}, \quad \chi_N = -\frac{m_N}{\Pi_N}, \quad (53)$$

and the exposure index

$$\mathcal{X}_j = \chi_j - \phi_b^j. \quad (54)$$

Around the normalized steady state ($s = \pi^ = 1$), with a binding EBC and holding quantities and prices fixed, an unexpected depreciation improves sector j 's net financing position if and only if $\mathcal{X}_j > 0$. The mechanical effect per unit of log depreciation is*

$$\theta_\pi \Pi_j \mathcal{X}_j.$$

In particular, $\chi_N < 0$, so $\mathcal{X}_N < 0$ for any $\phi_b^N \geq 0$.

The index makes operating and liability exposures commensurable: χ_j measures net dollar operating flow relative to earnings, while ϕ_b^j measures the dollar share of debt. Earnings-based borrowing maps both into the same financing margin. The index also has a direct covenant interpretation. With debt at face value $B_j = b_j^H + s b_j^F$, the elasticity of the debt-to-EBITDA ratio with respect to the exchange rate is

$$\frac{\partial \log(B_j/\Pi_j)}{\partial \log s} = \phi_b^j - \chi_j = -\mathcal{X}_j,$$

so a firm with $\mathcal{X}_j < 0$ that starts at its covenant limit is pushed above it by a depreciation unless it deleverages, while a firm with $\mathcal{X}_j > 0$ gains headroom. The result is mechanical rather than an exact equilibrium threshold; the multiplier also responds to interest rates, expected exchange rates, and dividend adjustment. The same experiment clarifies the roles of the borrowing constraint and invoicing currency.

Proposition 2 (When can a depreciation improve financing conditions?). *The direct effect of a depreciation on sector j 's net financing position is*

$$\theta_\pi \Pi_j (\chi_j - \phi_b^j) \quad \text{under the EBC}, \quad -\phi_b^j B_j \quad \text{under the debt-matched ABC}. \quad (55)$$

For exporters,

$$\chi_T^{DCP} = \frac{p^X y^X - m_T}{\Pi_T}, \quad \chi_T^{PCP} = -\frac{m_T}{\Pi_T} < 0, \quad (56)$$

while $\chi_N < 0$ under either pricing paradigm. Under DCP, $\chi_T^{DCP} > 0$ if and only if $p^X y^X > m_T$. Hence a depreciation can directly improve financing conditions for some admissible $\phi_b^j \geq 0$ only when the firm both borrows against earnings and has positive net dollar operating exposure.

Thus constraint type can change the *sign*, not only the magnitude, of the direct financial effect of depreciation. Under an ABC, lower dollar debt can only attenuate the effect; under an EBC, dollar revenues can overturn it. Dollar invoicing is also necessary for this reversal: under PCP, export revenues do not revalue with the exchange rate, so the exporter carries only the direct dollar input-cost exposure. Section 4.3.2 shows that this distinction also appears in general equilibrium. At the calibrated steady state,

$$\chi_T \simeq 0.78, \quad \chi_N \simeq -0.19.$$

The observed tradable dollar-debt share of 0.28 therefore leaves exporters in the hedged region, whereas non-tradables remain exposed even with zero dollar debt. For the tradable sector,

$$\chi_T = \frac{x_T - mc \alpha_m}{1 - mc(1 - \alpha)}, \quad x_T \equiv \frac{y^X}{y_T}, \quad (57)$$

so exposure rises with export intensity and falls with imported-input intensity. Across $x_T \in [0.30, 0.55]$ and $\alpha_m \in [0.05, 0.15]$, χ_T ranges from roughly 0.4 to above 1. When $\chi_T > 1$, even fully dollarized debt does not offset the firm's positive net dollar operating exposure. The relevant cutoff is therefore sector- and economy-specific rather than universal. Appendix D provides the derivation.

3.7 Steady state and calibration

We calibrate the model at a quarterly frequency. The deterministic steady state is analytic and is derived in Appendix E. We normalize steady-state inflation and relative prices to one, total hours to $N = 1/3$, and quarterly GDP to one.

Table 3 reports parameters common across sectors. The non-tradable expenditure weight is $\omega_N = 0.5$, yielding a non-tradable sector accounting for approximately one-half of steady-state domestic activity, and we set $\eta_N = 0.5$ following Stockman and Tesar (1995). The tradable-sector foreign-currency debt share is fixed at its empirical value, $\phi_b^T = 0.28$. For non-tradable firms, $\phi_b^N = 0.16$ is the empirical benchmark; we additionally consider $\phi_b^N = 0$ and $\phi_b^N = 0.75$.³

³The dollar shares are computed from public Capital IQ firms and the covenant multiple from covenanted syndicated loans, so the calibration describes firms at the constrained margin. Because the covenant is a cap rather than a target, an economy in which only a fraction of firms borrows against it scales the aggregate responses by that fraction while leaving the sectoral incidence unchanged. The

Parameter	Value	Description	Source or target
<i>Panel A. Preferences and demand</i>			
β	0.98	Discount factor	8% annual real rate
σ	2	Risk aversion	Mendoza (2010)
φ	1	Inverse Frisch elasticity	Frisch elasticity of one
ω_N	0.50	NT expenditure share	NT value-added share
η_N	0.50	T-NT elasticity	Stockman & Tesar (1995)
ω	0.30	Import share within tradables	Christiano et al. (2011)
η	1.50	Home-foreign elasticity	Gertler et al. (2007)
<i>Panel B. Technology and price setting</i>			
α	0.33	Capital share	Capital income share
α_m	0.10	Imported-input share	Mendoza (2010)
δ	0.025	Depreciation rate	10% per year
κ_I	4	Investment adjustment cost	Drechsel (2023)
ϵ	6	Elasticity across varieties	Markup of 1.2
κ_p	77	Price adjustment cost	Drechsel (2023)
σ_x	1.13	Foreign-demand elasticity	Gopinath et al. (2020)
<i>Panel C. Financing and balance sheets</i>			
ψ_d	0.46	Dividend adjustment cost	Jermann & Quadrini (2012)
τ	0.35	Corporate tax advantage	Jermann & Quadrini (2012)
θ_π	4×3.89	EBC tightness	Dealscan, Table 1
θ_k	≈ 0.99	ABC tightness	Matches EBC debt
ϕ_b^T	0.28	T dollar-debt share	Capital IQ, Table 2
ϕ_b^N	0.16	NT dollar-debt share	Capital IQ, Table 2
κ_D	0.001	NFA premium elasticity	Schmitt-Grohé & Uribe (2003)
$\bar{B}$	-1	Net foreign assets	-25% of annual GDP
<i>Panel D. Policy and shock processes</i>			
ϕ_π	1.50	Taylor-rule inflation resp.	De Leo et al. (2022)
ρ_R	0.75	Interest-rate smoothing	De Leo et al. (2022)
ρ_F	0.75	Foreign-rate persistence	De Leo et al. (2022)
ρ_z	0.90	Sectoral TFP persistence	Standard
ρ_{y^*}	0.80	Foreign demand, premium	Standard
ρ_τ	0.95	Trade-wedge persistence	Ambrosino et al. (2026)
$\sigma(\varepsilon^{r^F})$	0.0025	Foreign-rate shock s.d.	25 b.p.
<i>Panel E. Set to match steady-state targets</i>			
κ_L	—	Labor-disutility weight	$N = 1/3$
γ_z	—	Export-demand scale	$GDP = 1$

Table 3: Baseline quarterly calibration.

Note: The model is calibrated at quarterly frequency. θ_k , κ_L and γ_z are solved for jointly with the analytic steady state to hit the targets in the last column; all other parameters are set externally. The investment adjustment cost follows Smets and Wouters (2007) and Drechsel (2023). θ_π converts the Dealscan covenant limit of 3.89 times *annual* EBITDA into the quarterly constraint (15). Sectoral dollar-debt shares are the tradability split of Table 2; the counterfactual and stress values $\phi_b^N \in \{0, 0.75\}$ used below are not calibration targets.

The covenant convention deserves care, because the covenant caps the *face value* of debt at a multiple of *annual* EBITDA while the model is quarterly and Π_j is quarterly EBITDA. A binding steady-state constraint $B_j = \theta_\pi \Pi_j$ delivers a debt-to-annual-EBITDA ratio of $B_j/(4\Pi_j) = \theta_\pi/4$; matching the Dealscan loan-weighted mean of 3.89 therefore requires

$$\theta_\pi = 4 \times 3.89 = 15.56, \quad (58)$$

exactly — no discount factor appears, because neither the covenant nor (15) discounts the face value. The implementation uses this convention throughout, and the computed steady state satisfies

$$\frac{B_j}{4\Pi_j} = 3.89, \quad j \in \{T, N\}, \quad (59)$$

to machine precision. For the ABC economy, θ_k is chosen to reproduce the same steady-state debt. We set $\bar{B} = -1$, corresponding to net foreign liabilities of 25 percent of annual GDP. The resulting steady state has exports equal to approximately 26 percent of GDP. The implied sectoral allocation is reported in Appendix Table 5.

4 When does FX mismatch amplify shocks?

The experiments are organized by a single question: does FX mismatch amplify all depreciations, or only particular shocks? The design crosses three margins — *constraint type* (EBC versus debt-matched ABC), *source of the exchange-rate movement* (a foreign financial tightening versus trade shocks that move the exchange rate differently), and *sectoral FX exposure* (the tradable and non-tradable sectors' cash flows and dollar-debt shares) — with invoicing explaining why the tradable sector behaves differently from the non-tradable sector. We present the financial shock first, use the invoicing regime to establish what insures the exporter, then hold the depreciation fixed across shocks to show that its *source* determines the financial response, and close with trade fragmentation as an application.

4.1 A financial depreciation: sectoral incidence

Figure 2 traces the mechanism from operating exposure to financing conditions and investment. A 25bp foreign interest-rate tightening generates an impact nominal depreciation of 0.89 percent and a real depreciation of 0.72 percent, while CPI inflation rises by 0.17 percentage points. The exchange-rate, inflation, and policy responses are nearly identical across the three non-tradable dollarization calibrations. The cross-line differences in Figure 2 therefore primarily reflect differences in corporate FX exposure rather than differences in the aggregate shock.

results are also insensitive to the covenant median in place of the loan-weighted mean ($\theta_\pi = 4 \times 3.50$): the impact tightening of ζ_N is 1.33 against 1.47, and the non-tradable investment trough -0.35 against -0.38 .

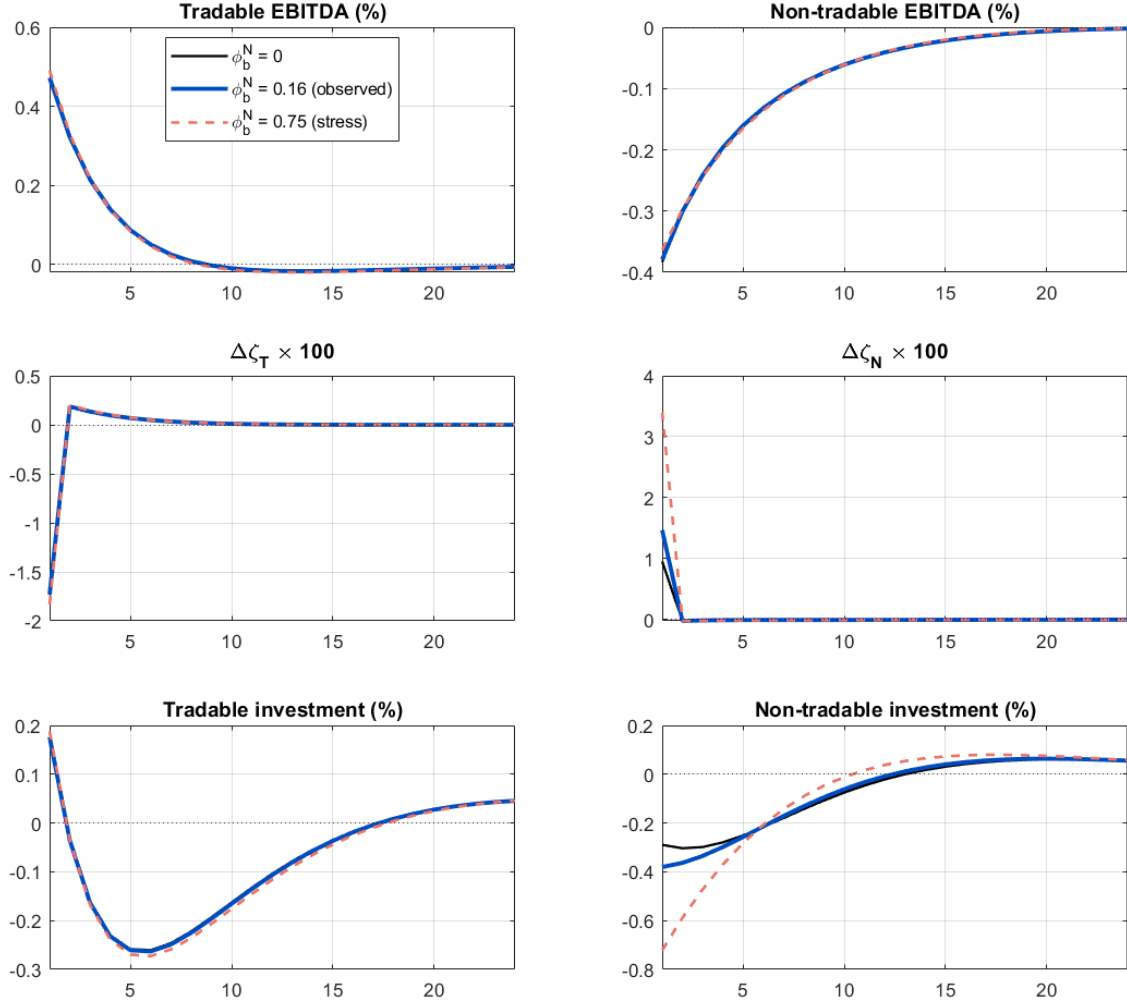


Figure 2: Sectoral transmission of a financial depreciation. Responses to a 25bp foreign interest-rate tightening at $\phi_b^N = 0$ (black), the observed $\phi_b^N = 0.16$ (thick blue), and the stress value $\phi_b^N = 0.75$ (dashed light red), with $\phi_b^T = 0.28$ throughout. Rows report EBITDA, the borrowing-constraint multiplier, and investment for the tradable and non-tradable sectors. EBITDA and investment are percent deviations from steady state; multiplier responses are deviations from steady state multiplied by 100. The impact nominal depreciation is 0.89 percent in all three experiments. Full macro and firm responses are reported in Appendix Figures 8–10.

The top row shows that the depreciation moves pledgeable earnings in opposite directions across sectors. Tradable EBITDA rises by 0.47 percent on impact: dollar export revenues are revalued into domestic currency while dollar-sticky export prices limit the immediate quantity response. Non-tradable EBITDA instead falls by 0.38 percent because the domestic-currency cost of imported inputs rises without an offsetting dollar revenue stream. Importantly, the three non-tradable EBITDA responses lie almost on top of one another. Changing the currency composition of debt therefore leaves the operating shock

essentially unchanged; it adds liability-side exposure to a common cash-flow shock. The middle row shows how earnings-based borrowing converts this operating exposure into financial exposure. Even with no dollar debt, the non-tradable multiplier rises by 0.96 on impact: lower EBITDA reduces the earnings against which the firm can borrow. Dollar liabilities then add a second source of exposure through the revaluation of inherited debt. At the observed dollar share, the impact increase in the multiplier rises to 1.47, and at $\phi_b^N = 0.75$ it reaches 3.40. The exporter moves in the opposite direction in the baseline calibration: its higher dollar earnings relax the constraint, so its multiplier falls on impact. Thus the same depreciation worsens financing conditions for the non-tradable firm while improving them for a sufficiently hedged exporter. The bottom row translates this financial incidence into investment. Non-tradable investment falls immediately by 0.30, 0.38, and 0.72 percent as ϕ_b^N rises from zero to 0.16 and 0.75. The increase in dollarization therefore deepens the investment contraction even though the underlying EBITDA loss is essentially unchanged. Tradable investment instead rises by about 0.18 percent on impact, consistent with the temporary improvement in earnings and financing conditions, before turning negative and reaching a trough of roughly -0.26 percent five to six quarters later. The initial financial hedge therefore affects the timing and incidence of investment, but does not fully insulate the tradable sector from the subsequent macroeconomic contraction. The central result is consequently visible in the separation between the rows of Figure 2: the operating shock to the non-tradable sector is nearly invariant to its debt currency, while its financing and investment responses become progressively more severe as dollar debt rises. Dollar liabilities amplify an exposure that already exists through dollar-denominated operating costs; they are not necessary for the financial tightening itself. Full macro and firm responses are reported in Appendix Figures 8–10. Because the linear solution pushes the exporter’s multiplier toward the zero boundary, we verify the result with occasionally binding constraints in Appendix F.4. In that solution the exporter becomes unconstrained for several quarters, whereas the non-tradable constraint remains binding at every dollarization level and its investment responses are virtually unchanged.

4.2 The exposure index in equilibrium

Proposition 1 characterizes the mechanical effect of a depreciation. Figure 3 asks whether the exposure index also organizes the full equilibrium. We vary each sector’s dollar-debt share continuously, holding the other sector at its observed value. For tradables, the impact response of the borrowing-constraint multiplier changes sign at $\phi_b^T \simeq 0.73$, close to the mechanical threshold $\chi_T = 0.78$. For non-tradables, the multiplier tightens at every admissible dollar-debt share, including zero, as implied by $\mathcal{X}_N < 0$. The index is therefore not an exact equilibrium cutoff: interest rates, expected exchange-rate dynamics, and payout adjustment also affect the multiplier. But it closely predicts the location of the tradable sign reversal and correctly orders exposure across sectors. In this sense, net dollar operating exposure pins down the direction of the direct financial effect exactly and is a strong predictor of the equilibrium response.

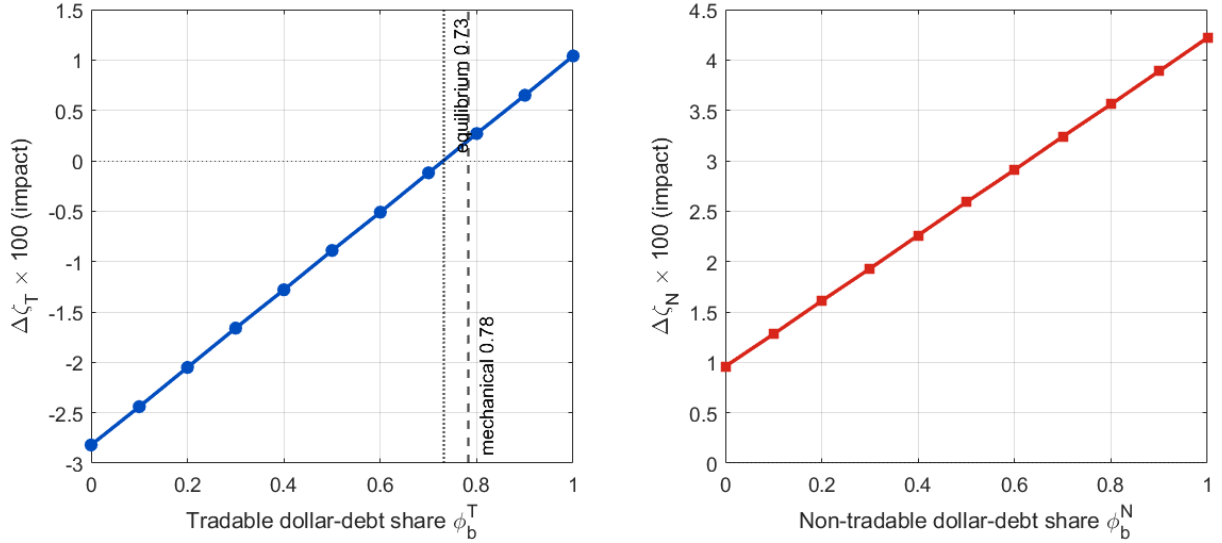


Figure 3: Exposure and equilibrium incidence. Impact response of each sector’s borrowing-constraint multiplier to the 25bp foreign interest-rate tightening as that sector’s dollar-debt share varies, holding the other sector at its observed value. For tradables, the equilibrium response changes sign at $\phi_b^T \simeq 0.73$, close to the mechanical threshold $\chi_T = 0.78$. For non-tradables, the multiplier tightens at every dollar-debt share, including zero, consistent with $\chi_N < 0$.

A qualification concerns covenant timing. The baseline constraint pledges contemporaneous quarterly earnings, whereas debt-to-EBITDA covenants are often written on trailing earnings. Under a four-quarter trailing EBITDA measure, a change in quarterly earnings enters the borrowing limit only gradually, which weakens both operating-cash-flow effects on impact. At the observed dollar-debt share the exporter becomes approximately neutral rather than positively insured. The non-tradable firm’s tightening at zero dollar debt weakens similarly: its multiplier dips on impact and rises over the following quarters, so that the effect approximately nets out over two years. The amplification from dollar debt, by contrast, is essentially unchanged, because the revaluation of inherited liabilities operates through the flow of funds rather than through the covenant window, and non-tradable investment still contracts at every dollar share. The component of the mechanism that is robust to covenant timing is therefore the liability channel and the sectoral asymmetry it produces; the operating-cash-flow channel, for exporters and non-exporters alike, is strongest when covenants track contemporaneous earnings or when depreciations persist relative to the covenant window. Appendix I reports the full trailing- EBITDA exercise.

4.3 Why does the sign change? Two mechanism counterfactuals

Proposition 2 shows that a depreciation can directly improve financing conditions only when firms borrow against earnings and have positive net dollar operating exposure. We

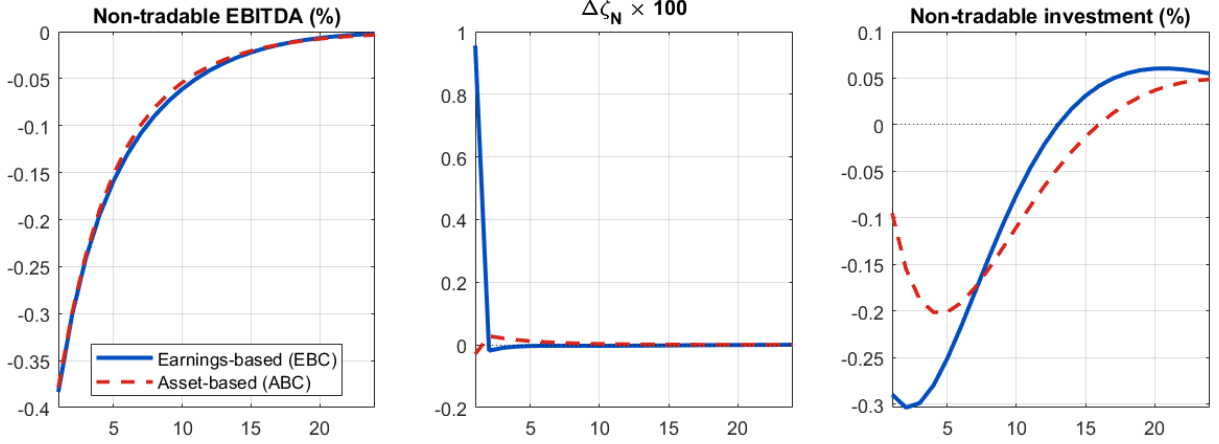


Figure 4: What firms pledge. Non-tradable responses to the 25bp foreign interest-rate tightening at $\phi_b^N = 0$ under the earnings-based constraint (solid) and the debt-matched asset-based constraint (dashed). The depreciation generates a similar operating cash-flow shock under the two constraints, but only the EBC maps that shock directly into borrowing capacity.

isolate these two ingredients in turn: first by replacing the earnings-based constraint with an asset-based constraint, and then by changing the invoicing currency.

4.3.1 What firms pledge: earnings versus assets

Figure 4 isolates the zero-dollar-debt result. We set $\phi_b^N = 0$ and compare the baseline earnings-based constraint with an asset-based constraint calibrated to match steady-state debt. The depreciation reduces non-tradable EBITDA by essentially the same amount under both constraints, about 0.38 percent. The operating exposure therefore does not depend on what the firm pledges. The financing response does. Under the EBC, lower EBITDA directly reduces pledgeable borrowing capacity and the non-tradable multiplier rises by 0.96 on impact. Under the ABC, imported-input costs still reduce earnings, but earnings are not collateral, so the multiplier is essentially unchanged (-0.03). Investment inherits this difference: the non-tradable trough is -0.30 percent under the EBC, compared with -0.20 percent under the ABC. The comparison separates the standard real cost of imported inputs from the financial amplification created by earnings pledgeability. A depreciation raises imported-input costs under either constraint, but this operating exposure becomes a borrowing-capacity shock only when firms borrow against earnings. The ABC is debt-matched rather than identical in every steady-state dimension, so we interpret the comparison as a mechanism counterfactual rather than an exact quantitative decomposition. Appendix H.1 reports the full ABC responses and additional comparisons.

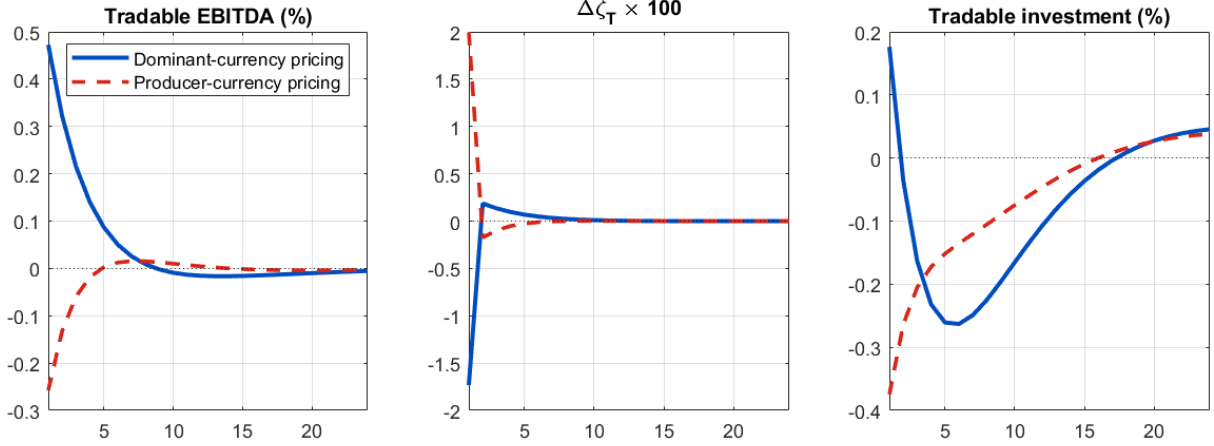


Figure 5: What currency firms invoice in. Tradable responses to the 25bp foreign interest-rate tightening under the earnings-based constraint at the observed calibration ($\phi_b^T = 0.28$, $\phi_b^N = 0.16$), under dollar-currency pricing (solid) and producer-currency pricing (dashed).

4.3.2 What currency firms invoice in: dollar versus producer pricing

Figure 5 isolates the role of invoicing currency by changing only the pricing convention. Under DCP, the depreciation raises the domestic-currency value of export revenues, so tradable EBITDA rises by 0.47 percent and the exporter’s borrowing constraint relaxes. Under PCP, by contrast, the depreciation stimulates export volumes through expenditure switching but does not mechanically revalue export revenues into domestic currency. Imported-input costs therefore dominate on impact: tradable EBITDA falls by 0.26 percent, the multiplier switches from -1.7 under DCP to $+2.0$ under PCP, and tradable investment turns from a 0.18 percent impact increase into a 0.38 percent contraction. The distinction is therefore between trade quantities and pledgeable earnings. PCP delivers the stronger export-volume response, but it is DCP that protects the constraint, because only dollar invoicing revalues the earnings against which the firm borrows. Together with Section 4.3.1, this completes the mechanism in Proposition 2: a depreciation can improve the exporter’s financing position only when firms borrow against earnings and those earnings have positive net dollar exposure. The broader incidence result does not hinge on the DCP assumption. Appendix H.3 shows that under an asset-based constraint the exporter’s multiplier moves little under either invoicing regime, while the non-tradable sector remains negatively exposed in both.

4.4 Same impact depreciation, different financial stress

The exposure index characterizes how a depreciation affects a firm conditional on its operating and liability positions. But the shock generating the depreciation also changes the earnings against which firms borrow. To isolate this distinction, we compare a foreign interest-rate tightening with an export-demand contraction, normalized to generate the same impact nominal depreciation. Their subsequent exchange-rate paths differ, which

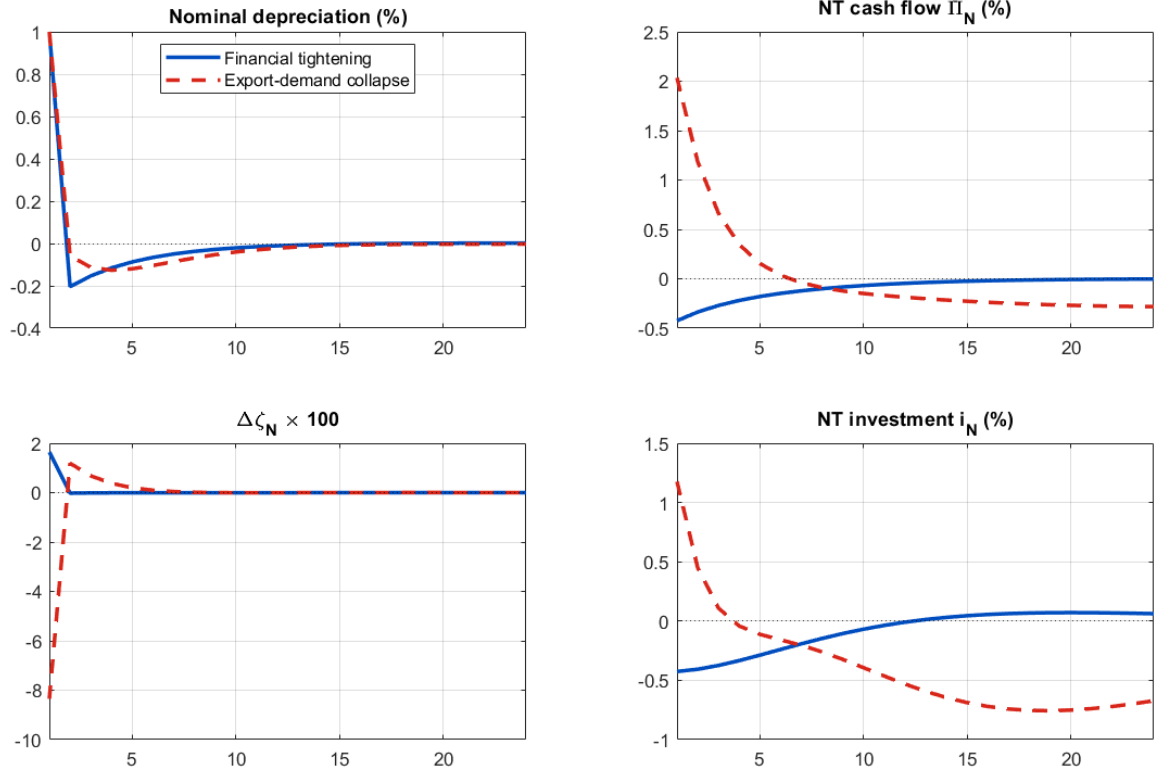


Figure 6: Same impact depreciation, different financial stress. Responses to a foreign interest-rate tightening (solid blue) and an export-demand contraction (dashed red), each normalized by its own impact nominal depreciation, at the observed calibration ($\phi_b^T = 0.28$, $\phi_b^N = 0.16$). The shocks generate the same impact depreciation but opposite responses of non-tradable EBITDA and financing conditions. Multiplier responses are from the linear solution; the occasionally-binding solution reported in Appendix J.3 makes the non-tradable constraint slack following the export-demand shock.

matters for equilibrium dynamics, but the normalization holds fixed the size and direction of the initial currency movement.

Figure 6 shows that the same impact depreciation need not imply the same financial stress. Following the financial tightening, non-tradable EBITDA falls by about 0.4 per cent per percent of depreciation and the borrowing constraint tightens. The depreciation raises imported-input costs and revalues inherited dollar liabilities while pledgeable earnings weaken. The export-demand contraction produces the opposite financing response. Although it also depreciates the currency and therefore raises the domestic value of dollar liabilities, it is deflationary: weaker tradable labor demand lowers the economy-wide wage and monetary policy eases. Non-tradable EBITDA consequently rises by about 2.0 per cent per percent of depreciation, more than offsetting the adverse liability revaluation. In the occasionally-binding solution the non-tradable constraint becomes slack rather than tightening. Investment reflects this difference in financing conditions on impact. The financial shock immediately contracts non-tradable investment, whereas the export-demand

shock initially raises it as the constraint relaxes. At longer horizons the adverse demand effect dominates and investment eventually falls under the export-demand shock as well. The distinction is therefore one of financial transmission, not of the eventual sign of the real contraction. The exchange rate alone is thus not a sufficient statistic for balance-sheet stress. What matters is the joint response of the exchange rate, pledgeable earnings, and foreign-currency liabilities. Identical balance sheets can experience opposite financial effects from depreciations of the same initial size because the shocks generating those depreciations move earnings differently. Appendix J decomposes the underlying cash-flow and liability movements and shows that the sign ranking is robust to alternative monetary-policy, price-stickiness, imported-input, and shock-persistence calibrations.

4.5 Trade fragmentation as an application

The previous section showed that a depreciation is not a sufficient statistic for financial stress. A delivered import-cost shock provides a complementary case: a shock can be strongly contractionary in real terms while generating little, or even favorable, FX balance-sheet pressure. We model fragmentation as a five-percent increase in the delivered price of imported intermediates, τ_t^m , with persistence 0.95. We refer to it as an import-cost or fragmentation shock rather than a tariff because the wedge accrues to foreign intermediaries; a domestic tariff with rebated revenue would introduce an additional fiscal transfer.

Figure 7 shows the resulting separation between real and financial effects. Higher imported-input costs reduce EBITDA and investment in both sectors, while the inflationary shock induces monetary tightening and an impact appreciation of about 1.1 percent. The appreciation lowers the domestic-currency value of inherited dollar liabilities. As a result, the non-tradable multiplier is approximately unchanged at the observed dollar share and falls under the stress calibration, even though non-tradable investment contracts sharply. Thus dollarization can provide balance-sheet relief at the same time that the underlying shock generates a severe real contraction. The relief is confined to impact, however: because the appreciation reverts, the dollarized firm services a heavier inherited obligation in later quarters, and its investment trough is the deeper one. Appendix K shows that this later effect runs entirely through the flow of funds — the marginal financing wedge is invariant to the dollar share beyond the impact quarter — so the episode separates the two channels the exposure index keeps distinct. The result reinforces the distinction between the direct cost of fragmentation and its financial amplification: imported-input costs are contractionary regardless of debt currency, whereas the contribution of dollar liabilities depends on the endogenous exchange-rate response. The sign of this financial component is not intrinsic to the fragmentation shock. It depends on monetary policy because policy helps determine the exchange-rate response. Appendix K shows that more accommodative rules can reduce or reverse the appreciation and thereby restore adverse balance-sheet effects; it also reports the two-sided fragmentation experiment and additional dollarization comparisons. Taken together with Section 4.4, the results imply that currency depreciations driven by financial shocks are especially likely to activate the

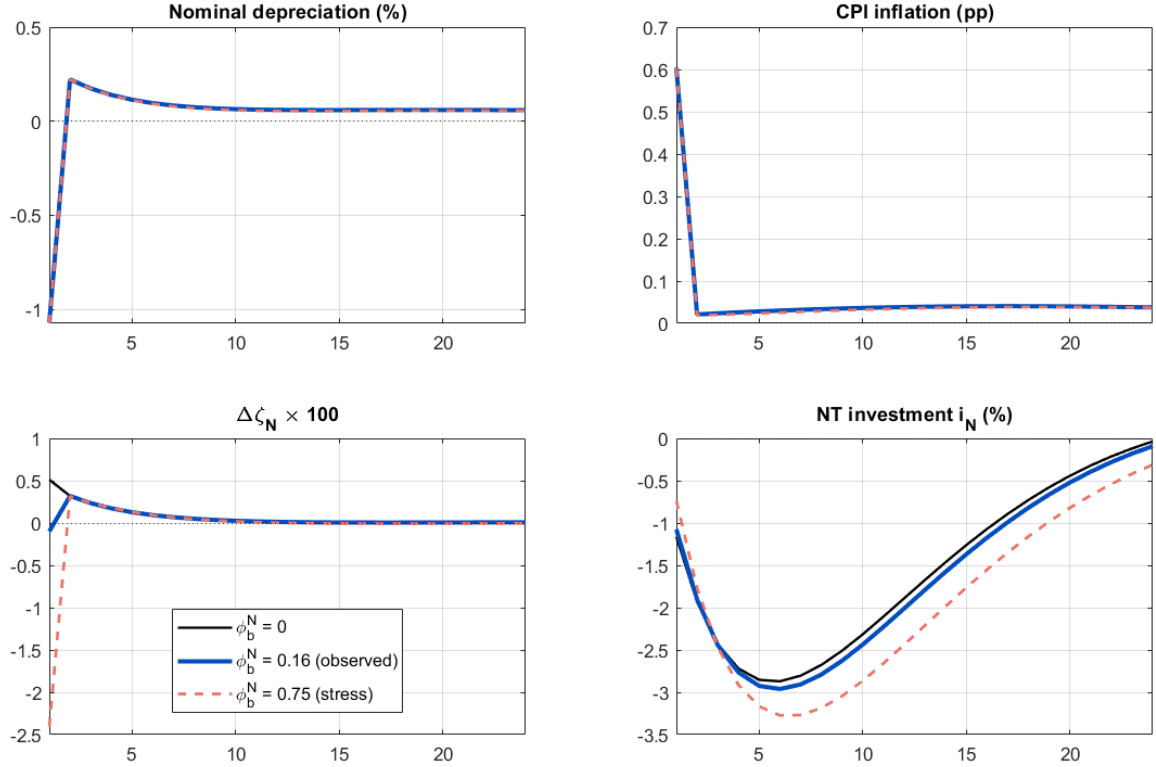


Figure 7: Trade fragmentation and financial incidence. Responses to a five-percent delivered import-cost shock at $\phi_b^N = 0$ (black), the observed $\phi_b^N = 0.16$ (thick blue), and the stress value $\phi_b^N = 0.75$ (dashed light red), with $\phi_b^T = 0.28$ throughout. The shock is strongly inflationary and contractionary, but the monetary response appreciates the currency on impact, reducing the domestic-currency burden of dollar liabilities.

FX-mismatch channel: they combine a weaker currency, weaker non-tradable earnings, and an adverse revaluation of dollar liabilities. Trade-driven exchange-rate movements need not produce the same configuration. This yields a cross-sectional empirical prediction: balance-sheet amplification should be stronger following currency depreciations driven by financial shocks, such as capital-flow reversals or external monetary tightenings, than following trade-driven exchange-rate movements of similar size.

5 Policy implications: balance-sheet versus demand stabilization

We conclude by illustrating how sectoral FX exposure changes monetary-policy transmission. The exercise is not a welfare comparison: evaluating policy with occasionally binding constraints would require a second-order treatment that is beyond the scope of the paper. Instead, we ask whether policies that stabilize the exposed sector's financing conditions also stabilize its real activity. The two-sector model generates a trade-off that is absent

Rule	$\mathcal{E}_0$ (%)	$\Delta\zeta_{N,0} \times 100$	i_N trough	y_N trough	$\sum i_N $	$\sum \pi $
Baseline ($\phi_\pi=1.5, \rho_R=0.75$)	0.89	3.40	-0.72	-0.086	4.24	0.21
No smoothing ($\rho_R=0$)	0.81	1.60	-0.68	-0.118	4.01	0.39
ER-augmented ($\phi_e=0.5$)	0.74	1.94	-0.60	-0.128	4.62	0.25
Hawkish ($\phi_\pi=3$)	0.77	2.41	-0.63	-0.118	4.82	0.17

Table 4: Monetary-policy transmission following a 25bp foreign tightening. Two-sector model with $\phi_b^N = 0.75$. Troughs are percent deviations from steady state. Cumulative columns report sums of absolute deviations over 40 quarters; $\sum |\pi|$ is in percentage points. These are descriptive statistics, not welfare losses.

from one-sector treatments. A stronger response to depreciation limits the revaluation of dollar liabilities and can therefore ease the non-tradable firm’s financing constraint. At the same time, the higher policy rate contracts domestic demand and changes both debt-service costs and the expected exchange-rate path. Balance-sheet stabilization therefore need not coincide with output stabilization. We compare four monetary rules following the same 25bp foreign tightening: the baseline Taylor rule ($\phi_\pi = 1.5$, $\rho_R = 0.75$), a rule without interest-rate smoothing, an exchange-rate-augmented rule ($\phi_e = 0.5$ on nominal depreciation), and a more inflation-responsive rule ($\phi_\pi = 3$). Table 4 reports the results for the stress calibration $\phi_b^N = 0.75$, where differences in balance-sheet exposure are most visible; the corresponding responses at the observed dollar share are smaller and reported in Appendix L. The exchange-rate-augmented rule illustrates the trade-off most clearly. Relative to the baseline, it reduces the impact depreciation from 0.89 to 0.74 percent and lowers the impact increase in the non-tradable multiplier from 3.40 to 1.94. The investment trough also becomes smaller, from -0.72 to -0.60 percent. Yet non-tradable output contracts more at its trough (-0.128 versus -0.086 percent), because the stronger policy response depresses domestic demand. Thus a rule that better stabilizes the sector’s financing position need not better stabilize its production. The remaining rules display the same lack of dominance. Removing interest-rate smoothing generates the smallest cumulative non-tradable investment and output deviations among the rules considered, but larger cumulative inflation deviations. The more inflation-responsive rule produces the smallest cumulative inflation deviation but larger cumulative real losses. These comparisons are descriptive rather than welfare rankings: the cumulative absolute deviations in Table 4 are diagnostic statistics, not policy loss functions. The broader implication is that sectoral FX mismatch changes the monetary-policy trade-off. Policies that limit depreciation can reduce balance-sheet stress for firms with negative FX exposure, but they do so by changing the same interest-rate and demand conditions that determine real activity. The relevant policy problem therefore depends not only on aggregate dollarization, but on which sectors hold the exposure and whether that exposure arises through liabilities or through the earnings firms pledge.

6 Conclusion

Earnings-based borrowing constraints turn the currency composition of *cash flows* into a determinant of financial amplification: what firms borrow against determines how the exchange rate reaches their borrowing capacity. Because dominant-currency invoicing puts exporters' revenues in dollars, their constraints carry natural insurance against depreciations when net dollar operating revenues are sufficiently large; non-tradable producers, short dollars in both flows and stocks, carry the mismatch — with or without dollar debt. And because the sign of the effect depends on the joint movement of the exchange rate, earnings, and dollar debt, a depreciation is not a sufficient statistic: a financial shock and an export-demand collapse generate comparable depreciations with opposite constraint responses, while a fragmentation shock — through the monetary response to its inflation — appreciates the currency, generating immediate valuation relief on dollar liabilities under an inflation-targeting rule, even though the reversion of that appreciation leaves dollarization mildly contractionary for investment through the debt-service flow. The mechanism also confronts monetary policy with a sectoral trade-off between exchange-rate and demand stabilization. The natural next steps are empirical — sector-level evidence on dollar debt and investment around financial versus trade shocks — and normative, characterizing optimal policy and macroprudential limits on non-tradable dollar borrowing in this environment.

References

- Aguiar, M. (2005). Investment, devaluation, and foreign currency exposure: The case of Mexico. *Journal of Development Economics*, 78(1), 95–113.
- Alfaro, L., Calani, M., & Varela, L. (2023). *Granular Corporate Hedging Under Dominant Currency* (NBER Working Paper No. 28910). National Bureau of Economic Research.
- Ambrosino, L., Chan, J., & Tenreyro, S. (2026). Trade fragmentation, inflationary pressures and monetary policy. *Journal of International Economics*.
- Amiti, M., Itskhoki, O., & Konings, J. (2022). Dominant currencies: How firms choose currency invoicing and why it matters. *The Quarterly Journal of Economics*, 137(3), 1435–1493.
- Bleakley, H., & Cowan, K. (2008). Corporate dollar debt and depreciations: Much ado about nothing? *The Review of Economics and Statistics*, 90(4), 612–626.
- Camara, S., & Sangiacomo, M. (2022). Borrowing constraints in emerging markets. *Working Paper*. <https://arxiv.org/abs/2211.10864>
- Casas, C., Meleshchuk, S., & Timmer, Y. (2023). The dominant currency financing channel of external adjustment. *IMF Working Paper Series*.
- Céspedes, L. F., Chang, R., & Velasco, A. (2004). Balance sheets and exchange rate policy. *American Economic Review*, 94(4), 1183–1193.
- Chava, S., & Roberts, M. R. (2008). How does financing impact investment? The role of debt covenants. *Journal of Finance*, 63(5), 2085–2121.
- Christiano, L. J., Trabandt, M., & Walentin, K. (2011). Introducing financial frictions and unemployment into a small open economy model. *Journal of Economic Dynamics and Control*, 35(12), 1999–2041.
- De Leo, P., Gopinath, G., & Kalemli-Özcan, Ş. (2022). *Monetary policy cyclicalities in emerging economies* (Working Paper No. 30458). National Bureau of Economic Research.
- Drechsel, T. (2023). Earnings-Based Borrowing Constraints and Macroeconomic Fluctuations. *American Economic Journal: Macroeconomics*, 15(2), 1–34.
- Drechsel, T., & Kim, S. (2024). Macroprudential policy with earnings-based borrowing constraints. *Journal of Monetary Economics*, 147, 103595.
- Galí, J., & Monacelli, T. (2005). Monetary policy and exchange rate volatility in a small open economy. *The Review of Economic Studies*, 72(3), 707–734.
- Galindo, A., Panizza, U., & Schiantarelli, F. (2003). Debt composition and balance sheet effects of currency depreciation: A summary of the micro evidence. *Emerging Markets Review*, 4(4), 330–339.
- Gertler, M., Gilchrist, S., & Natalucci, F. M. (2007). External constraints on monetary policy and the financial accelerator. *Journal of Money, Credit and Banking*, 39(2), 295–330.
- Gopinath, G., Boz, E., Casas, C., Díez, F. J., Gourinchas, P.-O., & Plagborg-Møller, M. (2020). Dominant Currency Paradigm. *American Economic Review*, 110(3), 677–719.

- Greenwald, D. L. (2019). *Firm Debt Covenants and the Macroeconomy: The Interest Coverage Channel* (Working Paper). MIT Sloan School of Management. https://www.dlgreenwald.com/uploads/4/5/2/8/45280895/debt_covenant_draft.pdf
- Hardy, B., & Saffie, F. (2024). From carry trades to trade credit: Financial intermediation by non-financial corporations. *Journal of International Economics*, 152, 103988.
- Hardy, B., Saffie, F., & Simonovska, I. (2026). Firm-to-firm financial linkages and dollar risk transmission. *Journal of Financial Economics*, 183.
- Jermann, U., & Quadrini, V. (2012). Macroeconomic effects of financial shocks. *American Economic Review*, 102(1), 238–271.
- Kim, J., & Lee, A. S. (2026). Liability dollarization and exchange rate pass-through to domestic prices. *Journal of International Economics*.
- Kim, J., Lee, S., & Lee, A. S. (2026). *Are exporters naturally hedged? corporate dollar debt and global trade* (Working Paper). Working paper.
- Kim, M. (2019). *Financial development, exchange rate fluctuations, and debt dollarization: A firm-level evidence* (IMF Working Paper No. 19/168). International Monetary Fund.
- Kim, Y. J., Tesar, L. L., & Zhang, J. (2015). The impact of foreign liabilities on small firms: Firm-level evidence from the Korean crisis. *Journal of International Economics*, 97(2), 209–230.
- Kiyotaki, N., & Moore, J. (1997). Credit cycles. *Journal of Political Economy*, 105(2), 211–248.
- Lian, C., & Ma, Y. (2020). Anatomy of Corporate Borrowing Constraints. *The Quarterly Journal of Economics*, 136(1), 229–291.
- Mendoza, E. G. (2010). Sudden stops, financial crises, and leverage. *American Economic Review*, 100(5), 1941–1966.
- Petersen, M. A., & Rajan, R. G. (1997). Trade credit: Theories and evidence. *Review of Financial Studies*, 10(3), 661–691.
- Roberts, M. R., & Sufi, A. (2009). Control rights and capital structure: An empirical investigation. *Journal of Finance*, 64(4), 1657–1695.
- Rotemberg, J. J. (1982). Monopolistic price adjustment and aggregate output. *The Review of Economic Studies*, 49(4), 517–531.
- Schmitt-Grohé, S., & Uribe, M. (2003). Closing small open economy models. *Journal of International Economics*, 61(1), 163–185.
- Smets, F., & Wouters, R. (2007). Shocks and frictions in US business cycles: A bayesian DSGE approach. *American Economic Review*, 97(3), 586–606.
- Stockman, A. C., & Tesar, L. L. (1995). Tastes and technology in a two-country model of the business cycle: Explaining international comovements. *American Economic Review*, 85(1), 168–185.

A Household optimality conditions

The household solves (28) subject to (29). The Lagrangian is

$$\begin{aligned} \mathcal{L}^h = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t & \left\{ \frac{C_t^{1-\sigma} - 1}{1-\sigma} - \kappa_L \frac{N_t^{1+\varphi}}{1+\varphi} \right. \\ & + \lambda_t \left[w_t N_t + \frac{a_{t-1}^H}{\pi_t} + \frac{s_t a_{t-1}^F}{\pi_t^*} + d_{T,t} + d_{N,t} \right. \\ & \left. \left. - C_t - \frac{a_t^H}{1+r_t^H} - \frac{s_t a_t^F}{(1+r_t^F)\Psi_t} - \mathcal{T}_t \right] \right\}. \end{aligned} \quad (60)$$

The first-order conditions are

$$[C_t] \quad \lambda_t = C_t^{-\sigma}, \quad (61)$$

$$[N_t] \quad \kappa_L N_t^\varphi = \lambda_t w_t, \quad (62)$$

$$[a_t^H] \quad 1 = \mathbb{E}_t \left[\Lambda_{t,t+1} \frac{1+r_t^H}{\pi_{t+1}} \right], \quad (63)$$

$$[a_t^F] \quad 1 = \mathbb{E}_t \left[\Lambda_{t,t+1} \Psi_t (1+r_t^F) \frac{s_{t+1}}{s_t \pi_{t+1}^*} \right], \quad (64)$$

where

$$\Lambda_{t,t+1} = \beta \frac{\lambda_{t+1}}{\lambda_t} = \beta \left(\frac{C_{t+1}}{C_t} \right)^{-\sigma}. \quad (65)$$

Combining the two bond Euler equations gives

$$\mathbb{E}_t \left[\Lambda_{t,t+1} \left(\frac{1+r_t^H}{\pi_{t+1}} - \Psi_t (1+r_t^F) \frac{s_{t+1}}{s_t \pi_{t+1}^*} \right) \right] = 0. \quad (66)$$

A.1 Intratemporal allocation

Cost minimization between tradable and non-tradable consumption is

$$\min_{C_{T,t}, C_{N,t}} \{ p_t^T C_{T,t} + p_t^N C_{N,t} \} \quad \text{s.t.} \quad C_t = \left[(1-\omega_N)^{1/\eta_N} C_{T,t}^{\frac{\eta_N-1}{\eta_N}} + \omega_N^{1/\eta_N} C_{N,t}^{\frac{\eta_N-1}{\eta_N}} \right]^{\frac{\eta_N}{\eta_N-1}}. \quad (67)$$

Since the CPI is the numeraire, cost minimization gives

$$C_{N,t} = \omega_N (p_t^N)^{-\eta_N} C_t, \quad C_{T,t} = (1-\omega_N) (p_t^T)^{-\eta_N} C_t, \quad (68)$$

together with

$$1 = (1-\omega_N) (p_t^T)^{1-\eta_N} + \omega_N (p_t^N)^{1-\eta_N}. \quad (69)$$

Within the tradable bundle,

$$\min_{C_{H,t}, C_{F,t}} \{p_t^H C_{H,t} + s_t C_{F,t}\} \quad \text{s.t.} \quad C_{T,t} = \left[(1-\omega)^{1/\eta} C_{H,t}^{\frac{\eta-1}{\eta}} + \omega^{1/\eta} C_{F,t}^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}}. \quad (70)$$

Hence,

$$C_{H,t} = (1-\omega) \left(\frac{p_t^H}{p_t^T} \right)^{-\eta} C_{T,t}, \quad (71)$$

$$C_{F,t} = \omega \left(\frac{s_t}{p_t^T} \right)^{-\eta} C_{T,t}, \quad (72)$$

and

$$(p_t^T)^{1-\eta} = (1-\omega)(p_t^H)^{1-\eta} + \omega s_t^{1-\eta}. \quad (73)$$

B Firm optimality conditions

B.1 Conditional cost minimization

For a firm i in sector $j \in \{T, N\}$, conditional on output and predetermined capital,

$$\min_{N_{j,i,t}, m_{j,i,t}} \{w_t N_{j,i,t} + s_t m_{j,i,t}\} \quad \text{s.t.} \quad y_{j,i,t} = z_{j,t} k_{j,i,t-1}^\alpha N_{j,i,t}^{1-\alpha-\alpha_m} m_{j,i,t}^{\alpha_m}. \quad (74)$$

The first-order conditions are

$$[N_{j,i,t}] \quad w_t = m c_{j,i,t} (1 - \alpha - \alpha_m) \frac{y_{j,i,t}}{N_{j,i,t}}, \quad (75)$$

$$[m_{j,i,t}] \quad s_t = m c_{j,i,t} \alpha_m \frac{y_{j,i,t}}{m_{j,i,t}}. \quad (76)$$

The marginal contribution of predetermined capital to operating earnings is

$$m c_{j,i,t} \alpha \frac{y_{j,i,t}}{k_{j,i,t-1}}. \quad (77)$$

Under the EBC, operating earnings enter the dynamic problem with coefficient

$$\mu_{j,i,t} + \tilde{\zeta}_{j,i,t} \theta_\pi = \mu_{j,i,t} (1 + \theta_\pi \zeta_{j,i,t}),$$

so the constraint does not distort the conditional labor-input mix.

B.2 Dynamic firm problem

Firm i in sector j maximizes

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \frac{\lambda_t}{\lambda_0} d_{j,i,t}. \quad (78)$$

Total face-value debt is

$$B_{j,i,t} = b_{j,i,t}^H + s_t b_{j,i,t}^F, \quad (79)$$

with

$$b_{j,i,t}^H = (1 - \phi_b^j) B_{j,i,t}, \quad s_t b_{j,i,t}^F = \phi_b^j B_{j,i,t}. \quad (80)$$

The flow-of-funds constraint is

$$\begin{aligned} d_{j,i,t} + \Phi(d_{j,i,t}) + i_{j,i,t} + AC_{j,i,t}^p + \frac{b_{j,i,t-1}^H}{\pi_t} + \frac{s_t b_{j,i,t-1}^F}{\pi_t^*} = \Pi_{j,i,t} + \frac{b_{j,i,t}^H}{1 + r_t^H(1 - \tau)} \\ + \frac{s_t b_{j,i,t}^F}{1 + [(1 + r_t^F)\Psi_t - 1](1 - \tau)}. \end{aligned} \quad (81)$$

The baseline borrowing constraint is

$$B_{j,i,t} \leq \theta_\pi \Pi_{j,i,t}. \quad (82)$$

Capital evolves according to

$$k_{j,i,t} = (1 - \delta)k_{j,i,t-1} + \left[1 - \frac{\kappa_I}{2} \left(\frac{i_{j,i,t}}{i_{j,i,t-1}} - 1 \right)^2 \right] i_{j,i,t}. \quad (83)$$

For the EBC economy,

$$\begin{aligned} \mathcal{L}_j^f = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \frac{\lambda_t}{\lambda_0} \left\{ d_{j,i,t} + \mu_{j,i,t} \left[\Pi_{j,i,t} + \frac{(1 - \phi_b^j) B_{j,i,t}}{1 + r_t^H(1 - \tau)} \right. \right. \\ + \frac{\phi_b^j B_{j,i,t}}{1 + [(1 + r_t^F)\Psi_t - 1](1 - \tau)} - d_{j,i,t} - \Phi(d_{j,i,t}) - i_{j,i,t} - AC_{j,i,t}^p \\ \left. \left. - \frac{(1 - \phi_b^j) B_{j,i,t-1}}{\pi_t} - \phi_b^j B_{j,i,t-1} \frac{s_t}{s_{t-1} \pi_t^*} \right] \right. \\ + \tilde{\zeta}_{j,i,t} [\theta_\pi \Pi_{j,i,t} - B_{j,i,t}] \\ \left. + \nu_{j,i,t} \left[(1 - \delta)k_{j,i,t-1} + \left(1 - \frac{\kappa_I}{2} \left(\frac{i_{j,i,t}}{i_{j,i,t-1}} - 1 \right)^2 \right) i_{j,i,t} - k_{j,i,t} \right] \right\}. \end{aligned} \quad (84)$$

$$q_{j,i,t} = \frac{\nu_{j,i,t}}{\mu_{j,i,t}}, \quad \zeta_{j,i,t} = \frac{\tilde{\zeta}_{j,i,t}}{\mu_{j,i,t}}. \quad (85)$$

The dividend condition is

$$1 = \mu_{j,i,t} [1 + \Phi'(d_{j,i,t})]. \quad (86)$$

Defining

$$\gamma_{j,i,t} = 1 + \Phi'(d_{j,i,t}), \quad (87)$$

gives $\mu_{j,i,t} = 1/\gamma_{j,i,t}$. In symmetric equilibrium,

$$M_{j,t,t+1}^f = \beta \frac{\lambda_{t+1}}{\lambda_t} \frac{\gamma_{j,t}}{\gamma_{j,t+1}} = \Lambda_{t,t+1} \frac{\gamma_{j,t}}{\gamma_{j,t+1}}. \quad (88)$$

Pricing equations:

$$\kappa_p(\pi_t^H - 1)\pi_t^H = (1 + \theta_\pi \zeta_{T,t}) \left(1 - \epsilon + \epsilon \frac{mc_{T,t}}{p_t^H} \right) + \mathbb{E}_t \left[M_{T,t,t+1}^f \kappa_p(\pi_{t+1}^H - 1)\pi_{t+1}^H \frac{p_{t+1}^H y_{t+1}^H}{p_t^H y_t^H} \right]. \quad (89)$$

$$\kappa_p(\pi_t^X - 1)\pi_t^X s_t = (1 + \theta_\pi \zeta_{T,t}) \left[(1 - \epsilon)s_t + \epsilon \frac{mc_{T,t}}{p_t^X} \right] + \mathbb{E}_t \left[M_{T,t,t+1}^f \kappa_p(\pi_{t+1}^X - 1)\pi_{t+1}^X s_{t+1} \frac{p_{t+1}^X y_{t+1}^X}{p_t^X y_t^X} \right]. \quad (90)$$

$$\kappa_p(\pi_t^N - 1)\pi_t^N = (1 + \theta_\pi \zeta_{N,t}) \left(1 - \epsilon + \epsilon \frac{mc_{N,t}}{p_t^N} \right) + \mathbb{E}_t \left[M_{N,t,t+1}^f \kappa_p(\pi_{t+1}^N - 1)\pi_{t+1}^N \frac{p_{t+1}^N y_{t+1}^N}{p_t^N y_t^N} \right]. \quad (91)$$

B.3 Capital, investment, and debt

Under the EBC, the capital condition is

$$q_{j,t} = \mathbb{E}_t \left[M_{j,t,t+1}^f \left\{ (1 + \theta_\pi \zeta_{j,t+1}) mc_{j,t+1} \alpha \frac{y_{j,t+1}}{k_{j,t}} + (1 - \delta) q_{j,t+1} \right\} \right]. \quad (92)$$

The investment condition is

$$1 = q_{j,t} \left[1 - \frac{\kappa_I}{2} \left(\frac{i_{j,t}}{i_{j,t-1}} - 1 \right)^2 - \kappa_I \left(\frac{i_{j,t}}{i_{j,t-1}} - 1 \right) \frac{i_{j,t}}{i_{j,t-1}} \right] + \mathbb{E}_t \left[M_{j,t,t+1}^f q_{j,t+1} \kappa_I \left(\frac{i_{j,t+1}}{i_{j,t}} - 1 \right) \left(\frac{i_{j,t+1}}{i_{j,t}} \right)^2 \right]. \quad (93)$$

The debt condition is

$$\begin{aligned} \zeta_{j,t} = (1 - \phi_b^j) & \left[\frac{1}{1 + r_t^H (1 - \tau)} - \mathbb{E}_t \left(M_{j,t,t+1}^f \frac{1}{\pi_{t+1}} \right) \right] \\ & + \phi_b^j \left[\frac{1}{1 + [(1 + r_t^F) \Psi_t - 1] (1 - \tau)} - \mathbb{E}_t \left(M_{j,t,t+1}^f \frac{s_{t+1}}{s_t \pi_{t+1}^*} \right) \right]. \end{aligned} \quad (94)$$

B.4 Asset-based constraint

For the ABC comparison,

$$B_{j,t} \leq \theta_k(1 - \delta)\mathbb{E}_t[q_{j,t+1}]k_{j,t}. \quad (95)$$

The individual firm takes the aggregate collateral price $q_{j,t+1}$ as given. The parameter θ_k is common across sectors and chosen so that steady-state debt matches the EBC economy's; because Π_j/k_j is identical across sectors in steady state, a single value reproduces both sectors' debt exactly. The capital condition becomes

$$q_{j,t} = \mathbb{E}_t \left[M_{j,t,t+1}^f \left\{ mc_{j,t+1} \alpha \frac{y_{j,t+1}}{k_{j,t}} + (1 - \delta)q_{j,t+1} \right\} \right] + \zeta_{j,t} \theta_k (1 - \delta) \mathbb{E}_t[q_{j,t+1}]. \quad (96)$$

C Aggregation and Walras' law

The aggregate household budget constraint is

$$C_t + \frac{a_t^H}{1 + r_t^H} + \frac{s_t a_t^F}{(1 + r_t^F)\Psi_t} + \mathcal{T}_t = w_t N_t + \frac{a_{t-1}^H}{\pi_t} + \frac{s_t a_{t-1}^F}{\pi_t^*} + d_{T,t} + d_{N,t}. \quad (97)$$

For each sector $j \in \{T, N\}$,

$$d_{j,t} + \Phi(d_{j,t}) + i_{j,t} + AC_{j,t}^p + \frac{b_{j,t-1}^H}{\pi_t} + \frac{s_t b_{j,t-1}^F}{\pi_t^*} = \Pi_{j,t} + \frac{b_{j,t}^H}{1 + r_t^H(1 - \tau)} + \frac{s_t b_{j,t}^F}{1 + [(1 + r_t^F)\Psi_t - 1](1 - \tau)}. \quad (98)$$

The government levies

$$\mathcal{T}_t = \sum_{j \in \{T, N\}} \left\{ \frac{b_{j,t}^H}{1 + r_t^H(1 - \tau)} - \frac{b_{j,t}^H}{1 + r_t^H} + \frac{s_t b_{j,t}^F}{1 + [(1 + r_t^F)\Psi_t - 1](1 - \tau)} - \frac{s_t b_{j,t}^F}{(1 + r_t^F)\Psi_t} \right\}. \quad (99)$$

that is, the tax exactly covers the difference between what firms raise under deductibility and what the same promises would raise at market rates. Consolidating the firms with the government therefore leaves the economy borrowing from abroad at market rates, which is why no after-tax discount factor survives below. Now sum (97), the two sector constraints (98), and the government budget (99). Dividends $d_{T,t} + d_{N,t}$ cancel between households and firms; the after-tax issuance terms appear once with each sign and cancel against (99); and domestic bond-market clearing, $a_t^H = b_{T,t}^H + b_{N,t}^H$, eliminates both the issuance and the repayment legs of home-currency debt. Collecting the remaining foreign positions with $\mathcal{B}_t = a_t^F - b_{T,t}^F - b_{N,t}^F$ gives the consolidated resource constraint

$$C_t + \sum_{j \in \{T, N\}} [i_{j,t} + \Phi(d_{j,t}) + AC_{j,t}^p] + \frac{s_t \mathcal{B}_t}{(1 + r_t^F)\Psi_t} - \frac{s_t \mathcal{B}_{t-1}}{\pi_t^*} = \Pi_{T,t} + \Pi_{N,t} + w_t N_t. \quad (100)$$

The left-hand side is domestic absorption plus net lending abroad. Writing absorption as

$$A_t = C_t + i_{T,t} + i_{N,t} + \sum_{j \in \{T,N\}} [\Phi(d_{j,t}) + AC_{j,t}^p], \quad (101)$$

and using the external constraint (51) to replace the two net-foreign-asset terms by the trade balance, $s_t \mathcal{B}_t / [(1 + r_t^F) \Psi_t] - s_t \mathcal{B}_{t-1} / \pi_t^* = TB_t$, equation (100) becomes $A_t + TB_t = \Pi_{T,t} + \Pi_{N,t} + w_t N_t$. Finally, the right-hand side is value added: the sectoral wage bills in (7)–(8) net out against $w_t N_t$ by labor-market clearing, leaving

$$\Pi_{T,t} + \Pi_{N,t} + w_t N_t = p_t^N y_t^N + p_t^H y_t^H + s_t p_t^X y_t^X - s_t (m_{T,t} + m_{N,t}) \equiv GDP_t, \quad (102)$$

and therefore

$$GDP_t = A_t + TB_t \quad (103)$$

The identity is not imposed as an equilibrium condition, so it provides an independent check on aggregation and market clearing: we compute the residual $GDP_t - A_t - TB_t$ along the simulated paths, and it is at machine precision in every experiment reported below. Under the trade-fragmentation experiment the same derivation goes through with imports valued at the delivered price s_t^m throughout — in import demand, in the sectoral input bills, and correspondingly in the trade balance (50) and value added (49), each of which replaces s_t by s_t^m on the import side. The wedge accrues to foreign intermediaries rather than to domestic agents, so it never enters the household or government budgets and passes through the external constraint unchanged; the identity continues to hold exactly. Both wedges are zero in the deterministic steady state, so the calibration of Section 3.7 is unaffected.

D Earnings-adjusted FX exposure

Consider an unexpected depreciation and hold quantities fixed. Under DCP,

$$\Pi_{T,t} = p_t^H y_t^H + s_t p_t^X y_t^X - w_t N_{T,t} - s_t m_{T,t}, \quad (104)$$

$$\Pi_{N,t} = p_t^N y_t^N - w_t N_{N,t} - s_t m_{N,t}. \quad (105)$$

Evaluated at the normalized steady state $s = 1$,

$$\frac{\partial \Pi_{T,t}}{\partial \log s_t} = p^X y^X - m_T, \quad (106)$$

$$\frac{\partial \Pi_{N,t}}{\partial \log s_t} = -m_N. \quad (107)$$

Under a binding EBC, $B_j = \theta_\pi \Pi_j$. Since $b_j^F = \phi_b^j B_j$ in the normalized steady state, the revaluation of inherited foreign-currency debt is

$$b_j^F = \phi_b^j \theta_\pi \Pi_j. \quad (108)$$

For the tradable sector, the net mechanical effect of depreciation is therefore

$$\theta_\pi(p^X y^X - m_T) - b_T^F = \theta_\pi \Pi_T \left[\frac{p^X y^X - m_T}{\Pi_T} - \phi_b^T \right]. \quad (109)$$

Define

$$\chi_T = \frac{p^X y^X - m_T}{\Pi_T}. \quad (110)$$

Then

$$\mathcal{X}_T = \chi_T - \phi_b^T. \quad (111)$$

For the non-tradable sector,

$$-\theta_\pi m_N - b_N^F = \theta_\pi \Pi_N \left[-\frac{m_N}{\Pi_N} - \phi_b^N \right], \quad (112)$$

so

$$\chi_N = -\frac{m_N}{\Pi_N} < 0, \quad \mathcal{X}_N = \chi_N - \phi_b^N < 0. \quad (113)$$

Hence dollar export revenues may hedge the tradable firm's foreign-currency liabilities, whereas the non-tradable firm is mechanically exposed through both imported inputs and foreign-currency debt. The indices are mechanical exposure statistics rather than exact equilibrium thresholds: the response of $\zeta_{j,t}$ also depends on expected exchange rates, interest rates, dividend smoothing, and the intertemporal dynamics of the financing problem. Proposition 2 follows from repeating this calculation in the two alternative environments. Under PCP the tradable firm sets a single domestic-currency price and sells both at home and abroad at that price, so $\Pi_{T,t} = p_t^H(y_t^H + y_t^X) - w_t N_{T,t} - s_t m_{T,t}$: the exchange rate now appears only in the input bill, and holding quantities and the producer-currency price fixed,

$$\frac{\partial \Pi_{T,t}}{\partial \log s_t} = -m_{T,t}, \quad \chi_T^{PCP} = -\frac{m_T}{\Pi_T} < 0. \quad (114)$$

The exporter's net dollar operating exposure therefore has the same sign as the non-tradable firm's, and $\mathcal{X}_T^{PCP} < 0$ at every $\phi_b^T \geq 0$: the hedge is a consequence of dollar invoicing, not of exporting per se. Under the ABC the pledged object is $\theta_k(1-\delta)\mathbb{E}_t[q_{j,t+1}]k_{j,t}$. In the mechanical experiment $k_{j,t}$ is predetermined and $q_{j,t+1}$ is held fixed along with all other prices and quantities, so $\partial(\text{pledgeable collateral})/\partial \log s_t = 0$; equilibrium movements in $q_{j,t+1}$ are an indirect general-equilibrium response, not a mechanical revaluation. The liability revaluation $-\phi_b^j B_j$ is identical across constraint types because it operates through the flow of funds, which the two economies share. Hence the direct effect under the ABC is $-\phi_b^j B_j \leq 0$.

E Analytical steady state

We characterize the deterministic steady state of the two-sector economy. Normalize

$$p^N = p^T = p^H = p^X = s = \pi = \pi^* = 1, \quad q_T = q_N = 1, \quad N = \frac{1}{3}, \quad GDP = 1,$$

	Tradable	Non-tradable
Gross output y_j	0.601	0.490
home / export	0.343 / 0.258	—
Employment N_j	0.184	0.150
Capital k_j	4.03	3.29
Investment i_j	0.101	0.082
Imported inputs m_j	0.050	0.041
EBITDA Π_j	0.265	0.216
Face-value debt $B_j = \theta_\pi \Pi_j$	4.13	3.37
Dollar-debt share ϕ_b^j	0.28	0.16
Net dollar operating exposure χ_j	0.78	-0.19

Table 5: Sectoral steady-state allocation.

Note: Deterministic steady state at the baseline calibration of Table 3, with $GDP = 1$, $C = 0.797$ and $TB = 0.02$. Quantities are in units of quarterly GDP. Both sectors satisfy the Dealscan covenant exactly, $B_j/(4\Pi_j) = 3.89$. Net dollar operating exposure is $\chi_T = (p^X y^X - m_T)/\Pi_T$ and $\chi_N = -m_N/\Pi_N$, as defined in Proposition 1; it is negative for the non-tradable sector at any dollar-debt share.

and set $\Psi = 1$. The household Euler equation implies

$$1 + \bar{r} = \frac{1}{\beta}. \quad (115)$$

Zero inflation and symmetric pricing imply

$$mc_T = mc_N \equiv mc = \frac{\epsilon - 1}{\epsilon}. \quad (116)$$

The steady-state borrowing multiplier is common across sectors:

$$\zeta = \frac{1}{1 + \bar{r}(1 - \tau)} - \beta. \quad (117)$$

Conditional factor demands imply

$$m_j = mc \alpha_m y_j, \quad w N_j = mc(1 - \alpha - \alpha_m) y_j, \quad j \in \{T, N\}. \quad (118)$$

Let

$$Y \equiv y_T + y_N, \quad y_T = y^H + y^X. \quad (119)$$

Since GDP is value added,

$$1 = Y - m_T - m_N = (1 - mc \alpha_m) Y, \quad (120)$$

and therefore

$$\boxed{Y = \frac{1}{1 - mc \alpha_m}.} \quad (121)$$

The steady-state external budget constraint gives

$$TB = -\frac{\bar{r}}{1 + \bar{r}}\bar{\mathcal{B}}. \quad (122)$$

Since $GDP = A + TB$,

$$A = 1 - TB. \quad (123)$$

At unit relative prices,

$$A_N = \omega_N A, \quad A_T = (1 - \omega_N)A, \quad (124)$$

$$A_H = (1 - \omega)(1 - \omega_N)A, \quad A_F = \omega(1 - \omega_N)A. \quad (125)$$

Goods-market clearing therefore gives

$$y_N = \omega_N A, \quad (126)$$

$$y_H = (1 - \omega)(1 - \omega_N)A, \quad (127)$$

$$y_X = Y - y_N - y_H. \quad (128)$$

With $y^* = p^X = 1$, the export-demand scale is

$$\gamma_z = y_X. \quad (129)$$

Summing labor demand across sectors yields

$$wN = mc(1 - \alpha - \alpha_m)Y. \quad (130)$$

Hence,

$$w = mc(1 - \alpha - \alpha_m)\frac{Y}{N}, \quad (131)$$

and sectoral employment is

$$N_j = N\frac{y_j}{Y}. \quad (132)$$

The sectoral capital condition gives

$$k_j = \frac{(1 + \theta_\pi \zeta)mc\alpha y_j}{1/\beta - 1 + \delta}, \quad i_j = \delta k_j. \quad (133)$$

Operating earnings are

$$\Pi_j = y_j - wN_j - m_j = [1 - mc(1 - \alpha)]y_j. \quad (134)$$

Because the EBC binds,

$$B_j = \theta_\pi \Pi_j. \quad (135)$$

Its currency composition is

$$b_j^H = (1 - \phi_b^j)B_j, \quad b_j^F = \phi_b^j B_j. \quad (136)$$

Thus, ϕ_b^T and ϕ_b^N change only the currency composition of steady-state debt and not real allocations.

Steady-state dividends satisfy

$$d_j = \Pi_j - i_j - B_j + \frac{B_j}{1 + \bar{r}(1 - \tau)}. \quad (137)$$

We set $\bar{d}_j = d_j$, so that $\Phi(d_j) = \Phi'(d_j) = 0$ and $\gamma_j = 1$ in steady state.

Aggregate consumption is

$$C = A - i_T - i_N. \quad (138)$$

The labor-disutility parameter is therefore

$$\kappa_L = \frac{C^{-\sigma} w}{N^\varphi}. \quad (139)$$

Finally, the household's foreign asset position is chosen so that aggregate net foreign assets equal the target:

$$a^F = \bar{\mathcal{B}} + b_T^F + b_N^F. \quad (140)$$

Given the normalization $GDP = 1$, the common steady-state productivity level is the value that satisfies

$$\bar{z} = \frac{y_j}{k_j^\alpha N_j^{1-\alpha-\alpha_m} m_j^{\alpha_m}}, \quad (141)$$

which is identical across sectors because k_j , N_j , and m_j are all proportional to y_j in steady state.

Since $\theta_\pi = 4 \times 3.89$,

$$\frac{B_j}{4\Pi_j} = 3.89, \quad j \in \{T, N\}. \quad (142)$$

The deterministic steady state is independent of (ϕ_b^T, ϕ_b^N) when domestic and foreign borrowing rates coincide, $\Psi = 1$, and the real exchange rate is constant.

F Full responses to the baseline financial tightening

This section reports the complete set of responses underlying Figure 2. The three experiments vary only the dollar share of non-tradable debt, $\phi_b^N \in \{0, 0.16, 0.75\}$, while holding the tradable share fixed at its observed value, $\phi_b^T = 0.28$. The exchange-rate, inflation, and monetary-policy responses are nearly identical across the three calibrations; the main differences therefore arise through corporate FX exposure.

F.1 Prices and monetary policy

The foreign tightening generates an impact nominal depreciation of 0.89 percent and a real depreciation of approximately 0.72 percent. CPI inflation rises by 0.17 percentage

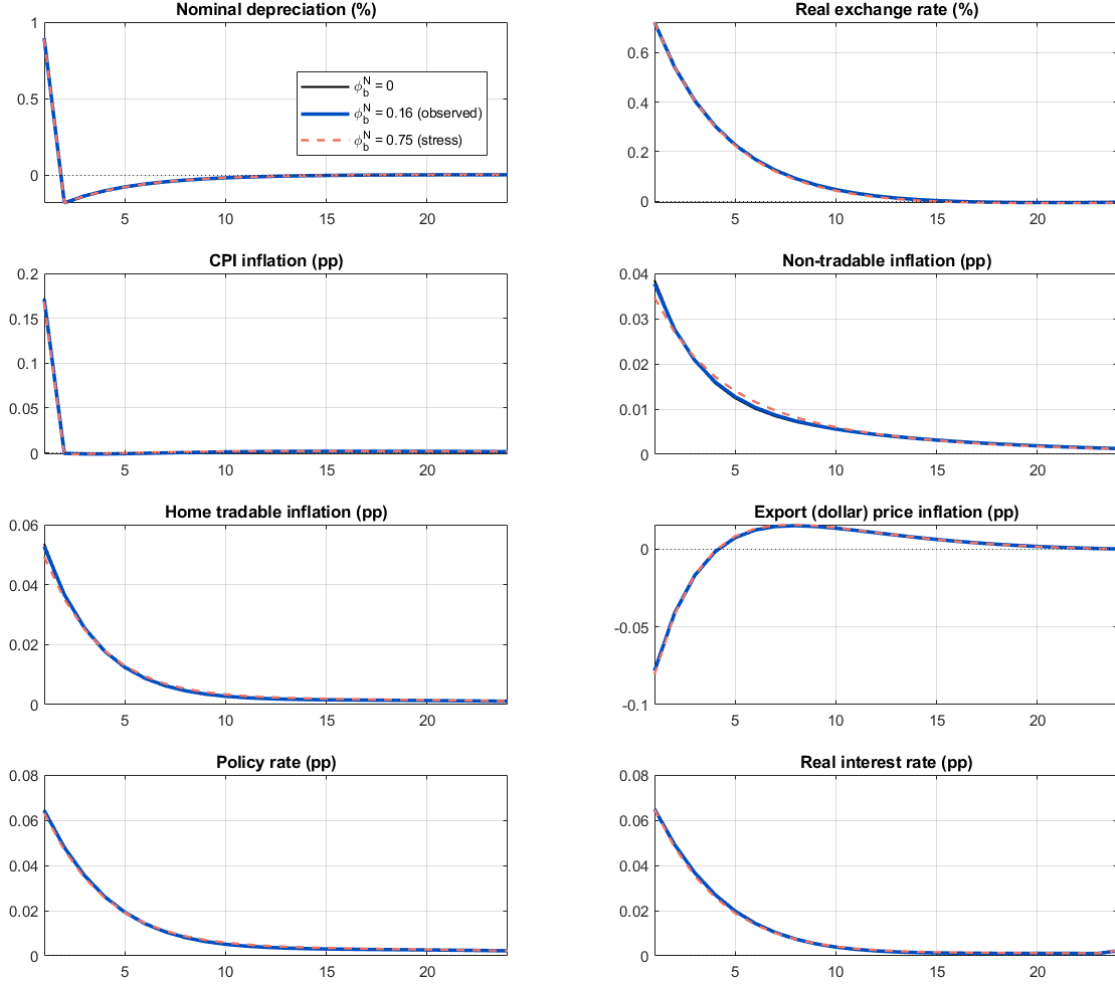


Figure 8: Baseline financial tightening: prices and monetary policy. Responses to a 25bp foreign interest-rate tightening at $\phi_b^N \in \{0, 0.16, 0.75\}$ and $\phi_b^T = 0.28$. Exchange rates are percent deviations from steady state; inflation and interest rates are in quarterly percentage points. The real interest rate is $r_t^H - \mathbb{E}_t \pi_{t+1}$. The near-coincidence of the three paths shows that changing non-tradable debt dollarization leaves the aggregate price and policy block nearly unchanged.

points on impact as imported prices pass through. Because the baseline Taylor rule features substantial interest-rate smoothing, $\rho_R = 0.75$, the domestic policy rate responds gradually and the real-rate response is modest on impact. These aggregate responses are essentially invariant to ϕ_b^N . The comparison across dollarization levels therefore does not operate by changing the initial exchange-rate or monetary-policy shock; instead, it changes how the same macroeconomic disturbance is transmitted through firm balance sheets and borrowing constraints.

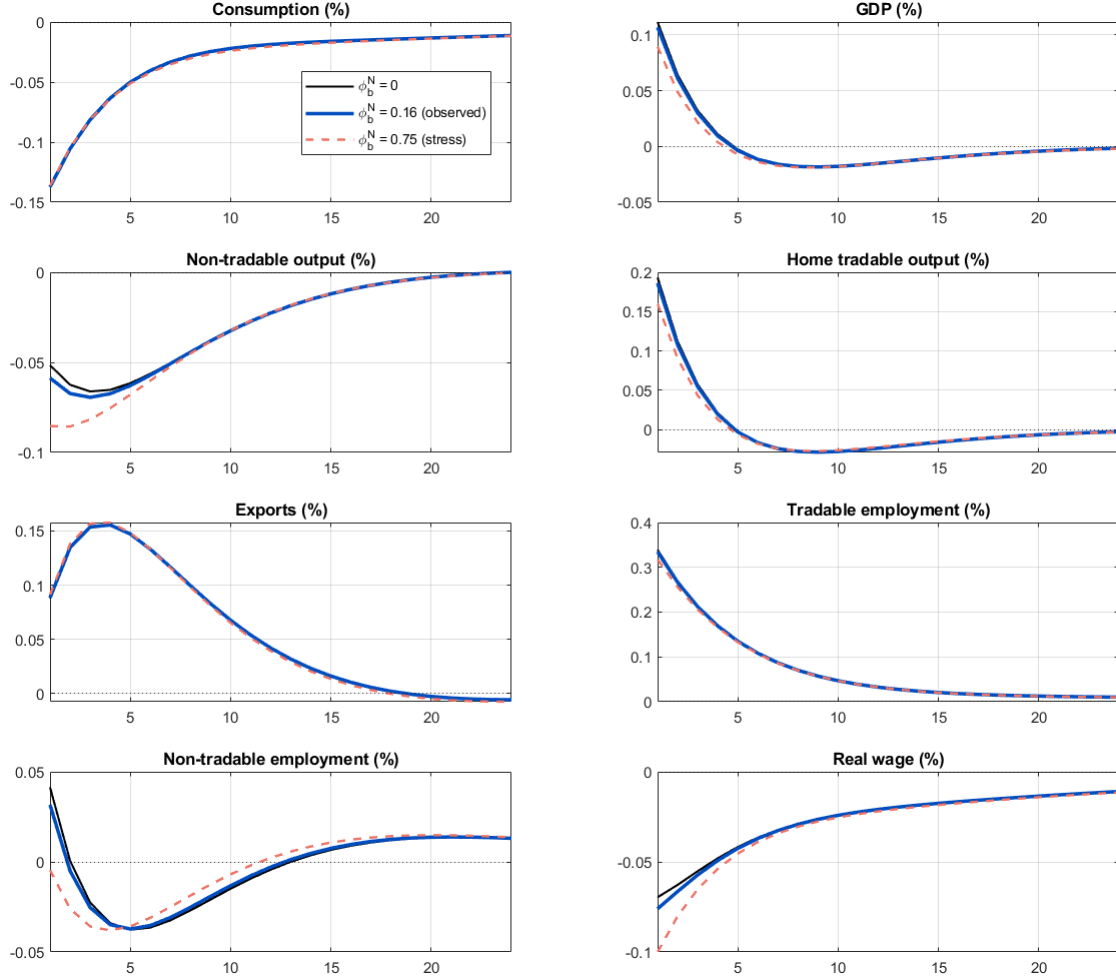


Figure 9: Baseline financial tightening: aggregate and sectoral quantities. Consumption, GDP, sectoral output, exports, employment, and the real wage following the 25bp foreign interest-rate tightening. All responses are percent deviations from steady state.

F.2 Aggregate and sectoral quantities

The depreciation initially reallocates activity toward the tradable sector. Exports and tradable production rise on impact, while non-tradable activity weakens. Aggregate GDP therefore moves little initially even though consumption falls. In particular, GDP rises about 0.11 percent on impact because expenditure switching toward domestic tradables partly offsets weaker domestic absorption. This response reflects the design of the experiment: the shock raises the foreign interest rate while holding foreign activity fixed. We therefore refer to it as a foreign interest-rate tightening rather than a complete U.S. monetary-policy shock, which would generally also move foreign demand. The sectoral composition of activity is more informative for the mechanism than aggregate GDP because the financial effects of the depreciation have opposite signs across sectors.

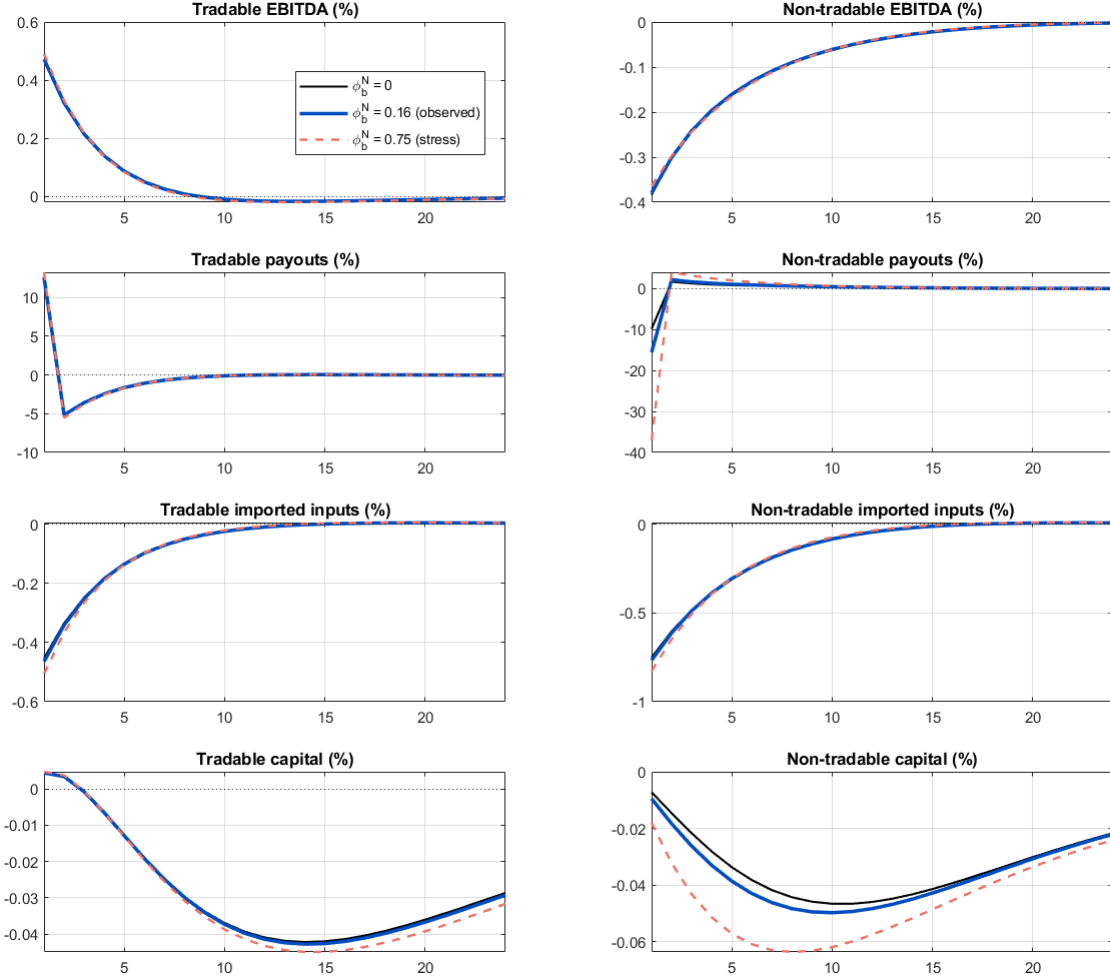


Figure 10: Baseline financial tightening: firm responses. Sectoral EBITDA, equity payouts, imported inputs, and capital stocks following the 25bp foreign interest-rate tightening. Responses are percent deviations from steady state.

F.3 Firm responses

The firm responses complete the transmission mechanism described in Figure 2. Tradable EBITDA rises on impact as dollar export revenues are revalued into domestic currency, while non-tradable EBITDA falls because imported-input costs increase without an offsetting dollar revenue stream. The EBITDA paths are almost unaffected by the currency composition of non-tradable debt. Firms also absorb part of the financing shock through payouts. Tradable payouts rise by approximately 12 percent on impact, whereas non-tradable payouts fall by roughly 10, 16, and 37 percent as ϕ_b^N increases from zero to 0.16 and 0.75. These percentage responses appear large relative to the sub-one-percent movements in EBITDA because payouts are only about 42 percent of steady-state EBITDA, whereas changes in earnings are mapped into borrowing capacity at the leverage multiple $\theta_\pi = 15.56$. At high dollarization, the revaluation of inherited dollar debt further

drains internal funds through the flow of funds. The capital-stock responses inherit the investment dynamics shown in Figure 2. Non-tradable capital falls progressively as the investment contraction accumulates, whereas the tradable capital response is delayed because tradable investment initially rises before subsequently contracting.

F.4 Occasionally binding constraints

The baseline linear solution is computed around a steady state in which both borrowing constraints bind. Because the foreign tightening lowers the tradable multiplier toward its zero boundary, we re-solve the model with a piecewise-linear solution in which each sectoral constraint can bind or become slack and $\zeta_j \geq 0$ is imposed. The regime-switching solution sharpens the baseline interpretation. Following the depreciation, the tradable constraint becomes slack and $\zeta_T = 0$ for several quarters. The non-tradable constraint, by contrast, remains strictly binding at every dollarization level, with $\zeta_N \geq 0.0067$ throughout the simulation. The non-tradable investment responses are virtually unchanged. The occasionally-binding solution produces troughs of -0.31 , -0.38 , and -0.72 percent for $\phi_b^N \in \{0, 0.16, 0.75\}$, compared with -0.30 , -0.38 , and -0.72 percent in the linear solution. Thus the sectoral incidence of the depreciation is not an artifact of linearizing around a binding steady state: economically, the shock temporarily unconstrains the exporter while tightening the non-tradable firm.

G Exchange-rate valuation and foreign funding costs

A foreign interest-rate tightening changes two objects simultaneously: the exchange rate and the dollar funding conditions faced by firms. This section separates these channels to establish whether the baseline dollarization result is primarily an exchange-rate valuation effect or a foreign-interest-rate effect.

G.1 Country-risk-premium shock

We first replace the foreign interest-rate shock with a shock to the country premium ξ_t in (31). This shock depreciates the currency without changing the foreign policy rate r_t^F . It is not a pure exchange-rate experiment because the premium Ψ_t also enters firms' effective foreign borrowing costs. Nevertheless, the cross-sectional dollarization effect is very similar to the baseline. Per percent of depreciation, increasing ϕ_b^N from zero to 0.75 deepens the non-tradable investment contraction by approximately 0.40 percentage points under the country-risk shock, compared with 0.47 under the foreign interest-rate shock. The sectoral incidence therefore does not depend on the shock entering specifically through the foreign policy rate.

G.2 Household-side UIP wedge

We next introduce a wedge in the household’s foreign-bond arbitrage condition that moves the exchange rate without entering firms’ foreign borrowing rate directly (TNT_DCP_EBC_UIPW). This experiment isolates the role of exchange-rate movements more tightly. The impact differential in the non-tradable multiplier between $\phi_b^N = 0.75$ and $\phi_b^N = 0$ is approximately 2.7 per percent of depreciation, essentially identical to the corresponding differential under the baseline financial shock. The investment differential is smaller, approximately -0.21 rather than -0.47 percent per percent of depreciation. Thus exchange-rate valuation accounts for nearly the full differential response of the borrowing constraint across dollarization levels, whereas movements in foreign funding costs contribute more materially to the propagation from financing conditions to investment.

G.3 Shutting down dollar-debt revaluation

As a final decomposition, we hold fixed the domestic-currency value of inherited dollar liabilities when the exchange rate moves (TNT_DCP_EBC_NOVAL). At the stress calibration, $\phi_b^N = 0.75$, the impact response of the non-tradable multiplier falls from 3.40 in the baseline to 1.34 when the revaluation of inherited dollar debt is removed. The comparison shows that the tightening associated with dollar liabilities is predominantly a valuation effect. The remaining response reflects the earnings-based flow mismatch and movements in foreign debt-service costs. Taken together, the three experiments show that the baseline financial-incidence result is not an artifact of interest-rate indexation: exchange-rate valuation generates most of the cross-sectional constraint response, while foreign funding costs help determine the subsequent investment response.

H Mechanism counterfactuals: additional results

This section reports the complete counterfactuals underlying Section 4.3. The first changes what firms pledge; the second changes the invoicing currency. The final subsection combines both dimensions.

H.1 Asset-based borrowing constraints

The ABC is calibrated to match sectoral steady-state debt in the EBC economy. Under the ABC, the non-tradable multiplier is essentially flat when $\phi_b^N = 0$: its impact response is -0.03 , compared with $+0.96$ under the EBC. Imported-input costs therefore remain an operating cost but no longer directly reduce pledgeable borrowing capacity. Dollarizing non-tradable debt still tightens the ABC through the revaluation of inherited foreign-currency liabilities. At $\phi_b^N = 0.75$, the multiplier rises by approximately 2.12. The investment response, however, is very different from the EBC: the non-tradable investment trough is approximately -0.20 percent with no dollar debt and -0.05 percent at the stress

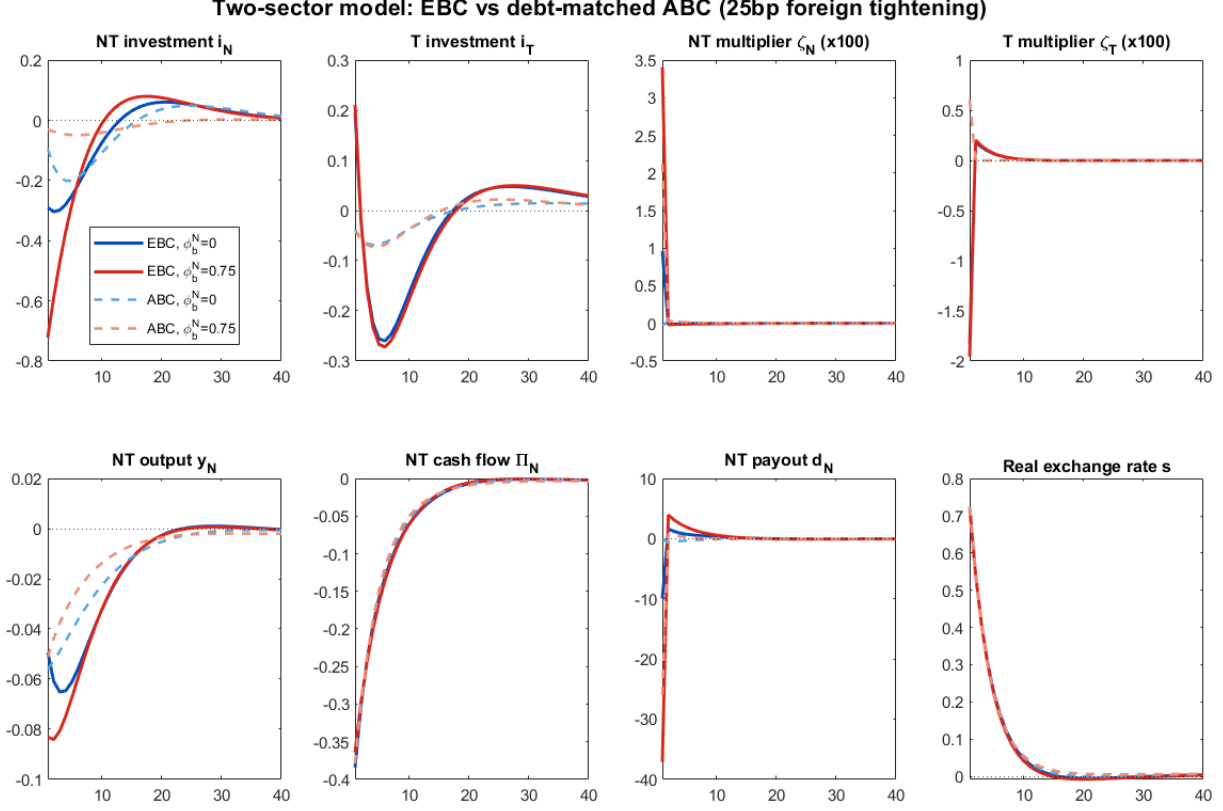


Figure 11: Earnings-based versus debt-matched asset-based borrowing constraints. Responses to the 25bp foreign interest-rate tightening under the EBC and ABC at alternative non-tradable dollar-debt shares.

dollarization, compared with -0.30 and -0.72 percent under the EBC. This apparently counterintuitive ABC response reflects the fact that capital itself is collateral. A tighter collateral constraint raises the shadow value of installed capital through the collateral-service term

$$\zeta_{N,t}\theta_k(1-\delta)\mathbb{E}_t q_{N,t+1}$$

in the capital condition. Constrained firms therefore have an additional incentive to preserve collateral precisely when the borrowing constraint tightens. This collateral-demand effect has no counterpart under the EBC, where earnings rather than capital are pledged. The ABC and EBC economies are debt-matched rather than identical in every steady-state dimension. In particular, steady-state capital is about six percent higher under the ABC because collateral service changes the user cost of capital. We therefore interpret the comparison as a mechanism counterfactual rather than an exact structural decomposition of the investment response.

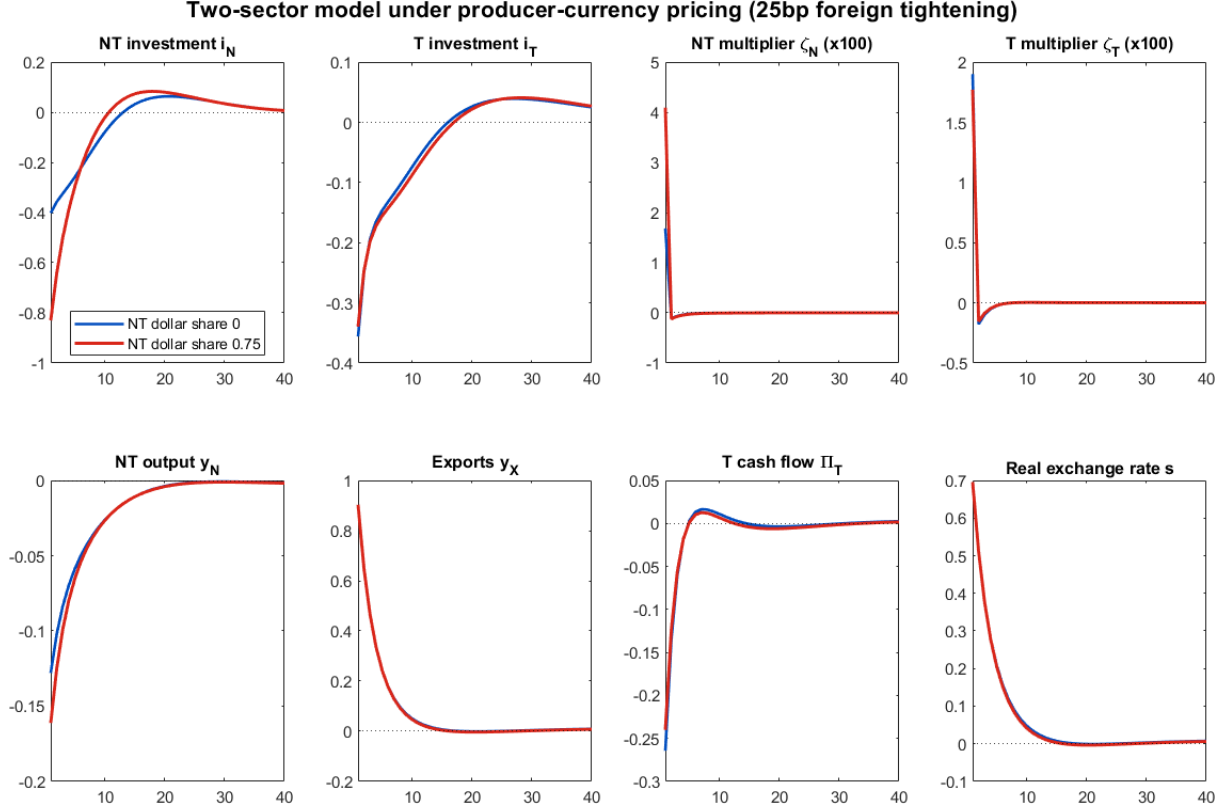


Figure 12: Producer-currency pricing. Responses to the 25bp foreign interest-rate tightening for $\phi_b^N \in \{0, 0.75\}$ and $\phi_b^T = 0.28$ under the earnings-based borrowing constraint.

H.2 Producer-currency pricing

Under producer-currency pricing, the depreciation lowers the foreign-currency price of exports and generates a strong export-volume response. It does not, however, mechanically increase the domestic value of dollar export revenues because export prices are set in producer currency. As a result, the tradable-sector financial response reverses relative to DCP. Tradable EBITDA falls on impact and the tradable multiplier rises by approximately 2.0, compared with a decline of about 1.7 under DCP. Tradable investment contracts immediately rather than rising on impact. The non-tradable result is qualitatively unchanged. Increasing ϕ_b^N from zero to 0.75 approximately doubles the non-tradable investment trough under PCP, from about -0.40 to -0.83 percent, compared with -0.30 to -0.72 percent under DCP. Thus the exporter hedge is specific to positive net dollar operating exposure, whereas the non-tradable flow-mismatch channel does not depend on the export-pricing paradigm.

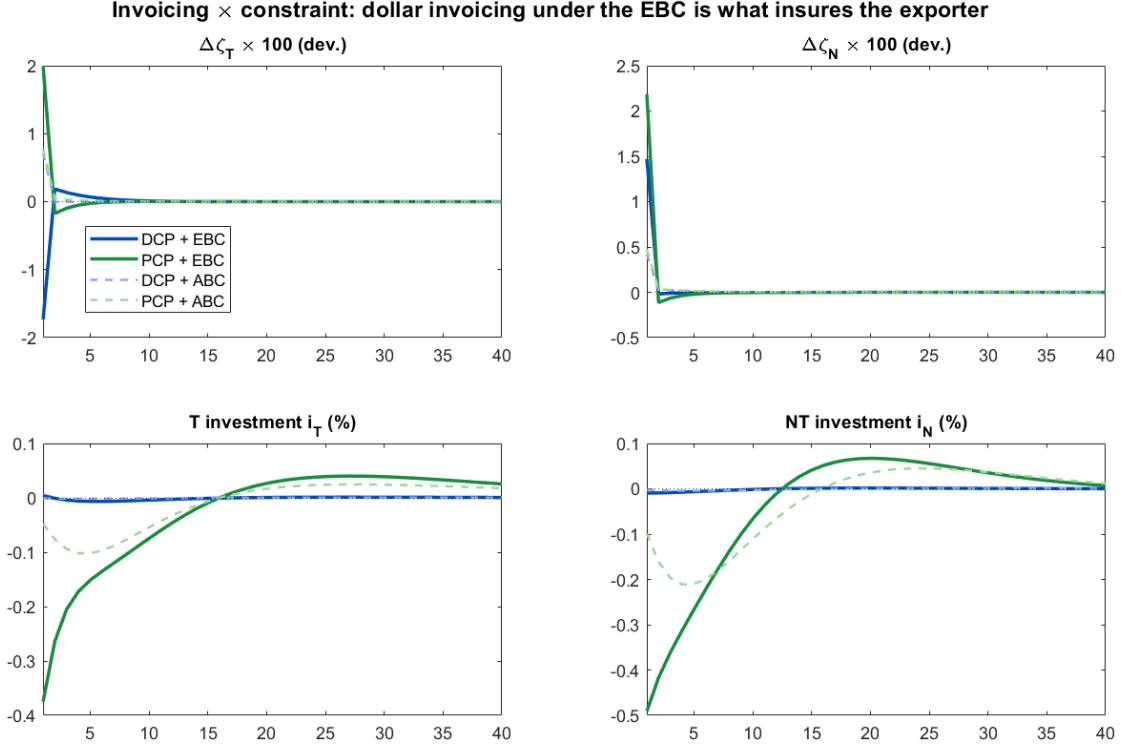


Figure 13: Constraint type $\times$ invoicing currency. Sectoral multiplier and investment responses to the 25bp foreign interest-rate tightening under DCP/PCP and EBC/ABC at the observed calibration ($\phi_b^T = 0.28$, $\phi_b^N = 0.16$).

H.3 Constraint type and invoicing currency

Figure 13 reports the full two-by-two comparison. Under the EBC, changing from DCP to PCP reverses the tradable multiplier response, from approximately -1.7 to $+2.0$. Under the ABC, by contrast, the tradable multiplier responds only modestly and similarly across pricing regimes, approximately $+0.72$ under DCP and $+0.78$ under PCP. The comparison summarizes the mechanism in Proposition 2. Earnings pledgeability makes operating currency exposure relevant for borrowing capacity, while dollar invoicing can make that operating exposure positive for exporters. Removing either ingredient eliminates the direct exporter sign reversal.

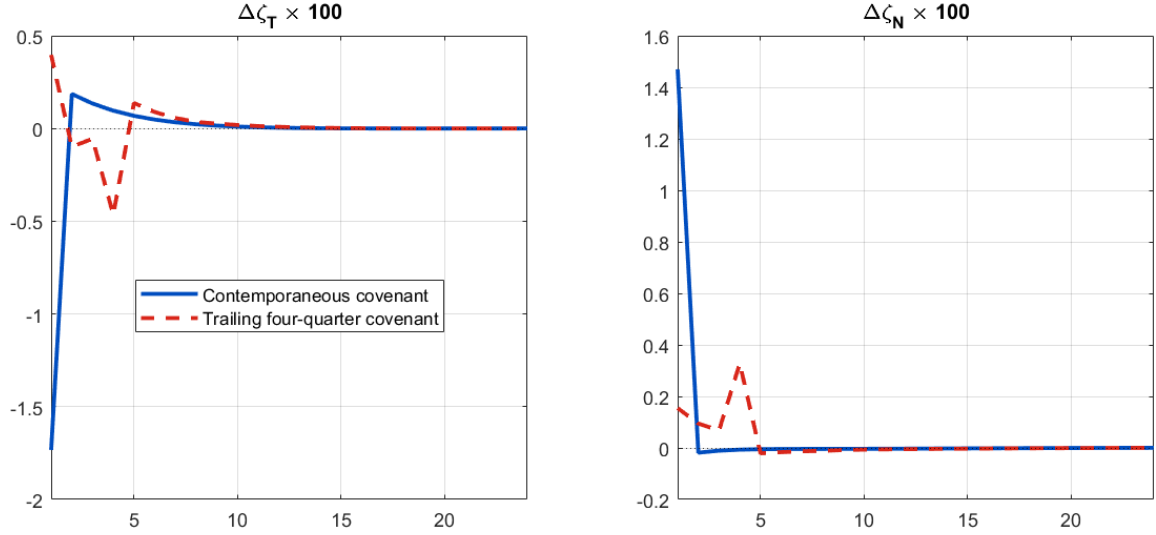


Figure 14: Covenant timing. Sectoral borrowing-constraint multipliers under the contemporaneous earnings constraint (solid) and a four-quarter trailing EBITDA constraint (dashed), following the 25bp foreign interest-rate tightening at the observed calibration.

I Covenant timing and effective FX exposure

I.1 Trailing EBITDA covenants

Actual debt-to-EBITDA covenants are commonly written using trailing rather than contemporaneous earnings. We therefore replace the baseline constraint with

$$B_{j,t} \leq \theta_\pi \bar{\Pi}_{j,t}, \quad \bar{\Pi}_{j,t} = \frac{1}{4} \sum_{k=0}^3 \Pi_{j,t-k},$$

and re-solve the model. The first-order-condition wedge on current earnings becomes the discounted contribution of current earnings to the next four covenant observations, and the steady state is recomputed accordingly.

The trailing covenant attenuates the impact sensitivity of borrowing capacity to current earnings. Under the contemporaneous constraint, a one-dollar change in quarterly earnings changes the debt limit by $\theta_\pi = 15.56$ dollars. Under the four-quarter trailing measure, the impact contribution of that observation is $\theta_\pi/4 = 3.89$ dollars, with the remainder entering as the observation persists in the covenant window. This timing adjustment primarily affects the exporter result. Under the baseline contemporaneous covenant, the equilibrium tradable sign reversal occurs at approximately $\phi_b^T = 0.73$. Under the trailing covenant it moves to approximately 0.19 on impact, close to the mechanical $\chi_T/4$ benchmark, and to approximately 0.26 when evaluated over a discounted eight-quarter horizon. At the observed tradable dollar share of 0.28, the exporter is therefore approximately neutral rather than strongly insured. The qualitative DCP/PCP ordering survives. Under PCP with a trailing covenant, the tradable multiplier remains substantially tighter

than under DCP and stays positive over horizons at which the DCP multiplier falls. The non-tradable results split in two. The component driven by dollar liabilities is robust: the multiplier rises monotonically with ϕ_b^N , and the spread in its impact response across dollarization levels is approximately 2.5, compared with 2.4 in the baseline, because the revaluation of inherited dollar liabilities operates through the flow of funds rather than through the timing of the covenant. The zero-dollar-debt component is not: at $\phi_b^N = 0$ the multiplier falls by about 0.4 on impact and rises only over the following quarters, so that its discounted sum over eight quarters is close to zero (about 0.05, against 0.91 under the contemporaneous covenant). Non-tradable investment nonetheless contracts at every dollarization level, with troughs somewhat deeper and later, at approximately -0.50 , -0.53 , and -0.67 percent. Covenant timing therefore matters for the operating-cash-flow channel in both sectors, turning the exporter hedge and the non-tradable flow tightening from impact results into persistence results, but not for the liability channel: dollar debt amplifies the non-tradable firm's exposure by essentially the same amount under either covenant.

I.2 Financial hedging

Observed dollar debt is a gross liability position rather than necessarily a firm's net FX exposure. As a reduced-form robustness exercise, let a fraction h of dollar liabilities be fully converted into domestic-currency exposure through a cross-currency hedge. The effective dollar share is then

$$\phi_{b,\text{eff}}^j = (1 - h)\phi_b^j.$$

At first order, this economy corresponds to moving leftward along the continuous exposure schedules in Figure 3. For example, hedge ratios $h \in \{0.25, 0.50, 0.75\}$ reduce the observed non-tradable dollar share of 0.16 to 0.12, 0.08, and 0.04, respectively. Because the non-tradable multiplier tightens even at $\phi_b^N = 0$, financial hedging of dollar *debt* cannot eliminate the operating-flow exposure generated by imported inputs. Complete hedging of the debt stock removes the liability revaluation channel but leaves the zero-dollar-debt tightening intact. The equivalence applies to a hedge that converts both principal and debt-service exposure into domestic-currency terms, such as a full cross-currency swap. A simple FX forward on principal would leave foreign-interest-rate and country-premium exposure in place and is not covered by this mapping.

J Shock dependence: decomposition and robustness

This section provides additional evidence for Section 4.4. We first compare the impact financing positions generated by the four shocks, then verify the export-demand result with occasionally binding constraints and alternative calibrations.

	Financial tightening	Country risk	Export demand	Import cost
Real exchange rate	+0.81	+0.81	+1.15	−1.56
CPI inflation (pp)	+0.19	+0.19	−0.15	+0.56
Tradable EBITDA	+0.53	+0.54	+1.04	−1.19
Non-tradable EBITDA	−0.42	−0.40	+2.03	−0.78
$\Delta\zeta_T \times 100$	−1.94	−1.98	−5.89	+2.11
$\Delta\zeta_N \times 100$	+1.64	+1.49	−8.34	−0.09
Tradable investment	−0.29	−0.33	−0.46	−1.80
Non-tradable investment	−0.43	−0.45	−0.76	−2.77

Table 6: Common diagnostic set across shocks. Responses are divided by the absolute impact nominal exchange-rate movement; EBITDA and investment are percent deviations from steady state and inflation is in percentage points. The two-sector EBC model is evaluated at the observed calibration, $\phi_b^T = 0.28$ and $\phi_b^N = 0.16$. Multiplier entries are from the linear solution and should be interpreted together with the occasionally-binding results below when a constraint reaches its zero boundary.

Shock	Flows (EBITDA components)				Capacity	Debt		
	$\Delta\text{Rev.}$	$-\Delta\text{Wages}$	$-\Delta\text{Inputs}$	$\Delta\Pi_N$	$\theta_\pi\Delta\Pi_N$	USD reval.	LC erosion	$\Delta\zeta_N$
Financial tightening	−0.11	+0.01	+0.00	−0.09	−1.43	−0.43	+0.54	+1.64
Country-risk premium	−0.11	+0.02	+0.00	−0.09	−1.36	−0.43	+0.55	+1.49
Export-demand collapse	+0.02	+0.36	+0.06	+0.44	+6.85	−0.61	−0.44	−8.35
Import cost	−0.56	+0.33	+0.06	−0.17	−2.62	+0.83	+1.60	−0.09

Table 7: Anatomy of the non-tradable firm’s financing position by shock. Entries are impact responses per one percent of absolute nominal exchange-rate movement, expressed in percent of steady-state GDP; a positive entry improves the firm’s financing position. Flow columns decompose $\Delta\Pi_N$ into revenue, wage-bill, and imported-input contributions. “Capacity” is the corresponding change in pledgeable borrowing capacity, $\theta_\pi\Delta\Pi_N$. “USD reval.” denotes the relief (+) or drain (−) from revaluation of inherited dollar debt; “LC erosion” denotes the real repayment relief from inflation on home-currency debt. The final column reports the linearized constraint multiplier response and is not the sum of the preceding columns.

J.1 A common diagnostic set across shocks

J.2 Financing-position decomposition

Table 7 decomposes the impact movement in the non-tradable firm’s financing position into operating-flow and liability components. The columns are not an exact accounting decomposition of $\Delta\zeta_N$, which also depends on discounting, expected exchange rates, interest rates, and payout adjustment. Their purpose is to identify which financing margins distinguish the shocks.

Under the financial tightening and the country-risk shock, non-tradable EBITDA falls while inherited dollar liabilities are revalued adversely. The loss of pledgeable earnings

and the liability revaluation therefore work in the same direction. The export-demand contraction is different. Tradable labor demand falls, reducing the economy-wide wage. The resulting decline in the non-tradable wage bill raises EBITDA sufficiently to dominate the adverse revaluation of dollar liabilities. Earnings pledgeability magnifies this otherwise modest cost decline into a large increase in borrowing capacity. The import-cost shock combines a negative capacity effect with favorable debt revaluation. The direct increase in delivered input costs reduces earnings, but the impact appreciation lowers the local-currency burden of dollar debt and inflation erodes home-currency repayments.

J.3 OBCs after the export-demand shock

The linear solution produces large negative deviations of both borrowing- constraint multipliers following the export-demand contraction. We therefore re-solve the model allowing the constraints to become slack. Both sectors reach the zero-multiplier regime. The tradable constraint becomes exactly slack, while the non-tradable multiplier reaches zero up to the numerical switch-point tolerance. The non-tradable investment trough is approximately -0.40 percent under the occasionally-binding solution, compared with -0.39 percent in the corresponding linear solution. Thus the negative linear multiplier response should not be interpreted as an economically negative Kuhn–Tucker multiplier. The correct interpretation is that the export-demand contraction temporarily unconstrains the firm. The absence of financial amplification under the export-demand shock is therefore a statement about constraint regimes, not a linearization artifact.

J.4 Parameter robustness

The sign contrast between financial and export-demand shocks is robust to substantial variation in parameters governing monetary policy, nominal rigidity, imported-input intensity, and shock persistence. We vary the Taylor coefficient over $\phi_\pi \in \{1.2, 3\}$, the Rotemberg parameter over $\kappa_p \in \{40, 120\}$, the imported-input share over $\alpha_m \in \{0.05, 0.15\}$, and the persistence of foreign demand over $\rho_{y^*} \in \{0.6, 0.9\}$. Across these configurations, the financial shock always tightens the non-tradable constraint, with responses between approximately $+1.0$ and $+2.1$ per percent of depreciation. The export-demand contraction always relaxes the constraint, with the linearized response ranging from approximately -3.9 to -20.9 per percent of depreciation. Both shocks continue to depreciate the currency in every configuration. The exact multiplier magnitudes are therefore parameter-dependent, but the ranking underlying Section 4.4 is not.

J.5 Financial amplification across shocks

The two financial shocks generate similar dollarization amplification, supporting the interpretation that the result is not specific to the foreign policy-rate component of the baseline experiment. The export-demand shock generates essentially no dollarization amplification: pledgeable earnings rise sufficiently to offset the adverse revaluation of dollar

Shock	Impact FX (%, + = dep.)	Dollarization differential	
		observed $\phi_b^N = 0.16$	stress $\phi_b^N = 0.75$
Foreign rate	+0.89	−0.09	−0.47
Country-risk premium	+4.37	−0.05	−0.40
Export demand	+0.52	+0.01	+0.03
Import cost	−1.07	−0.09	−0.38

Table 8: Dollarization amplification across shocks. The last two columns report the difference between the non-tradable investment trough at the indicated dollar share and the $\phi_b^N = 0$ counterfactual, divided by the absolute impact nominal exchange-rate movement. Negative values indicate that dollarization deepens the contraction. The second column reports the direction and size of the exchange-rate movement.

Taylor rule	$\mathcal{E}_0$ (%)	CPI ₀ (pp)	$\Delta\zeta_{N,0} \times 100$	
			$\phi_b^N = 0.16$	$\phi_b^N = 0.75$
Near-accommodative ($\phi_\pi = 1.1$)	−0.37	+0.93	+2.93	+1.67
Mild ($\phi_\pi = 1.2$)	−0.69	+0.78	+1.56	−0.20
Baseline ($\phi_\pi = 1.5$)	−1.07	+0.60	−0.10	−2.41
Hawkish ($\phi_\pi = 3.0$)	−1.54	+0.43	−2.63	−6.02

Table 9: Monetary policy and the financial incidence of fragmentation. Impact responses to the five-percent delivered import-cost shock under Taylor rules differing only in ϕ_π , with $\rho_R = 0.75$ throughout. A negative $\mathcal{E}_0$ denotes an appreciation.

liabilities. The fragmentation shock is more subtle. Its impact balance-sheet effect is stabilizing because the currency appreciates, yet its eventual investment differential remains non-negligible as the exchange rate subsequently returns toward steady state and financing conditions evolve. Even so, the dollarization differential is small relative to the direct contraction generated by the import-cost shock itself.

K Trade fragmentation: additional results

K.1 Monetary-policy dependence

The balance-sheet relief generated by the baseline fragmentation shock is not intrinsic to the trade-cost disturbance. It arises because the inflationary shock induces monetary tightening and an impact appreciation. We therefore vary the inflation coefficient in the Taylor rule while holding the remaining parameters fixed.

Under the baseline rule, the currency appreciates by approximately 1.1 percent and the non-tradable constraint is approximately unchanged at the observed dollar share and relaxes at the stress calibration. Under the near-accommodative rule, the appreciation is much smaller, inflation rises more strongly, and the non-tradable constraint tightens.

A more aggressive inflation response instead produces a larger appreciation and stronger balance-sheet relief. The financial incidence of fragmentation therefore depends on the endogenous exchange-rate response, which in turn depends on monetary policy. This policy dependence motivates the separate policy experiments in Section 5.

K.2 Why dollarization deepens the fragmentation trough

Table 8 contains an apparent tension: under the import-cost shock a higher dollar share *loosens* the non-tradable constraint on impact, yet the investment trough is *deeper* at the higher dollar share. Reconstructing both legs of the financing wedge (20) from the simulated paths resolves it, and the resolution is not the one intuition suggests. From the second quarter onward, $\zeta_{N,t}$ is invariant to the dollar share to five decimal places: the home-currency and dollar legs move by almost exactly offsetting amounts (at $t = 2$, -0.0059 and $+0.0059$ between the observed and stress calibrations), so the marginal financing wedge is the same in the two economies. Its discounted sum over twenty quarters is, if anything, *lower* at high dollarization (-0.014 against $+0.008$). The deeper trough is therefore not a constraint-tightening effect. It is a flow-of-funds effect. The impact appreciation is temporary: the real exchange rate returns toward its steady state from below as the wedge decays, so from $t = 2$ the dollarized firm services a heavier inherited dollar obligation in *every* subsequent quarter than the less dollarized firm. The resulting drain on internal funds is small per period but persistent, and with the marginal wedge unchanged it must be absorbed by payouts and investment: non-tradable payouts run 1 to 2 percentage points lower at the stress calibration from $t = 2$ onward, while sectoral EBITDA is essentially identical across the two economies. The capital stock consequently crosses over — higher at the stress calibration for the first three quarters, lower from the fifth — and investment inherits the same crossover. This is the *level* channel of Section 3, the revaluation of inherited liabilities in the flow of funds, operating without any change in the marginal channel that the exposure index describes.

K.3 Two-sided fragmentation

The main text isolates the delivered import-cost shock because it maps directly into the non-tradable flow mismatch. We also consider a two-sided fragmentation experiment that combines the five-percent import wedge with a five-percent foreign tariff on domestic exports.

The combined shock is more strongly stagflationary than the pure import-cost experiment. CPI inflation rises by approximately 0.6 percentage points on impact and GDP falls by about 1.8 percent at its trough. Nevertheless, the FX balance-sheet component remains relatively small compared with the direct trade-cost effect, and increasing non-tradable dollarization changes the investment contraction only modestly. The two-sided experiment therefore reinforces the main fragmentation result while combining two distinct mechanisms—higher imported-input costs and weaker export demand. For transparency, the main text uses the pure import-cost experiment to isolate the former.

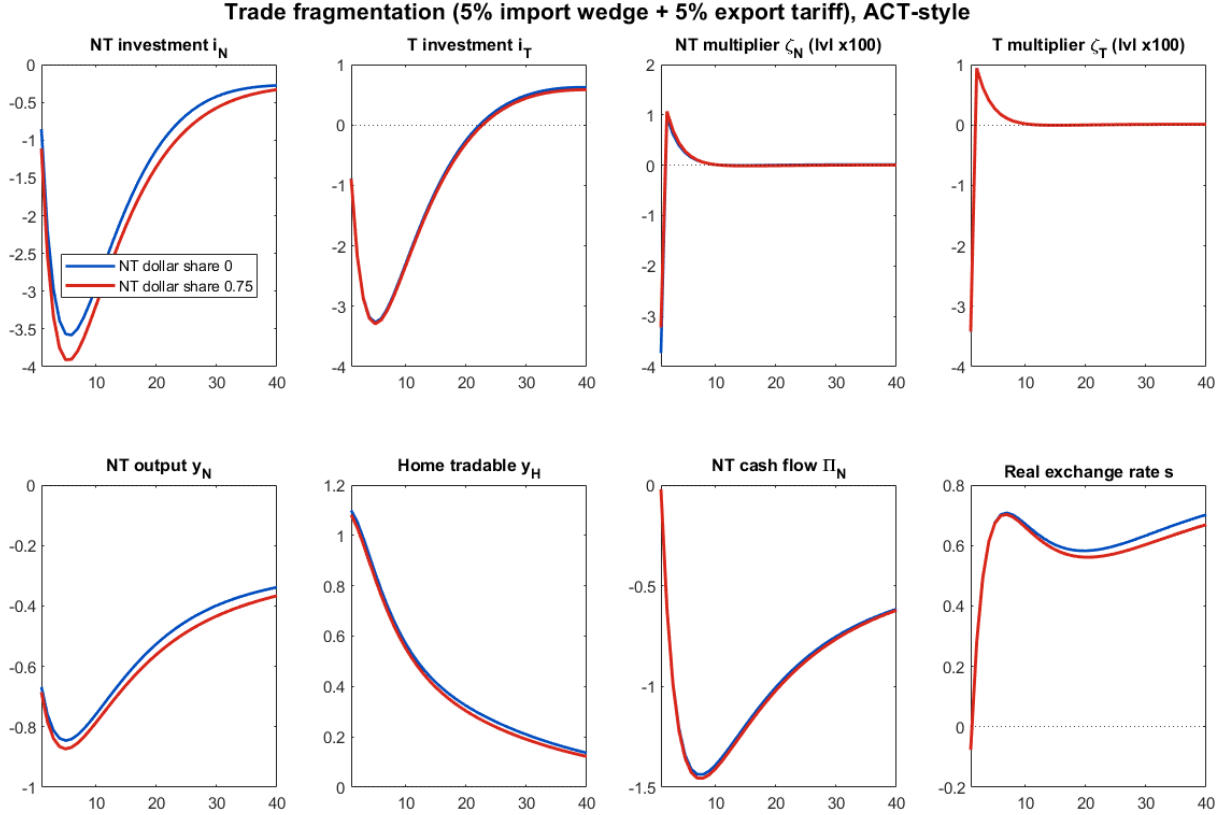


Figure 15: Two-sided fragmentation. Responses to a five-percent delivered import-price wedge combined with a five-percent foreign export tariff, shown at alternative non-tradable dollarization levels.

L Monetary-policy rules: full responses

Figure 16 reports the full impulse responses underlying the policy comparison in Section 5. The experiment fixes the foreign interest-rate shock and the stress non-tradable dollarization, $\phi_b^N = 0.75$, and varies the monetary-policy rule.

The exchange-rate-augmented rule produces the smallest impact depreciation among the rules considered and substantially reduces the tightening of the non-tradable borrowing constraint. The accompanying interest-rate response, however, depresses domestic demand more strongly, so improved balance-sheet stabilization does not imply smaller output losses. Removing interest-rate smoothing front-loads the policy response and generates comparatively small cumulative real deviations, but larger cumulative inflation deviations. The more inflation-responsive rule stabilizes inflation more effectively while producing larger cumulative real losses. These comparisons are descriptive transmission exercises, not welfare rankings.

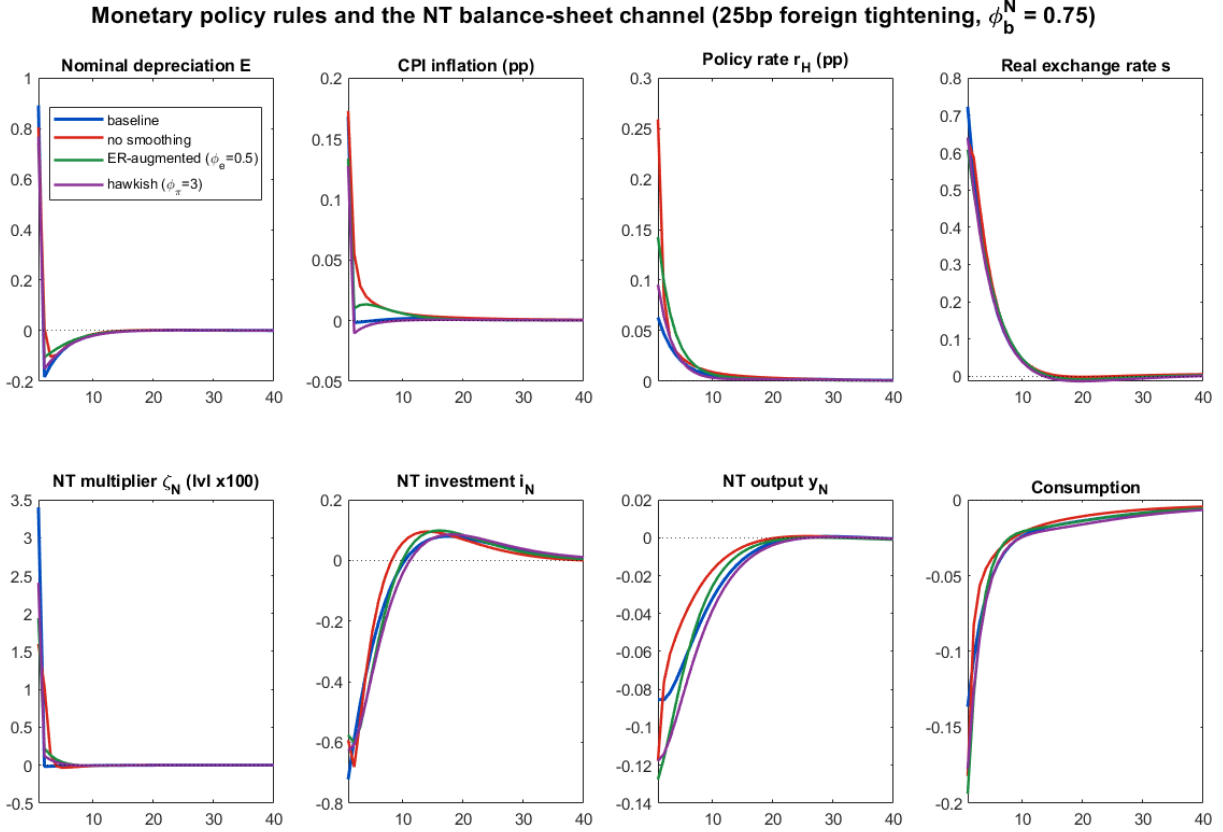


Figure 16: Monetary-policy rules and the non-tradable balance-sheet channel. Responses to the 25bp foreign interest-rate tightening under the baseline, no-smoothing, exchange-rate-augmented, and hawkish rules of Table 4; $\phi_b^N = 0.75$.

M Data and empirical details

Capital IQ. Debt denomination is measured at the instrument level for 10,292 public non-financial firms in 29 EM countries (headquarters basis), cleaned following Kim (2019). Figure 17 reports the dollar share of debt by country-year and by industry-year, and its evolution over time: dollar borrowing is broad-based across industries, including non-tradable ones — the fact the two-sector model is built to interpret.

Limitations of the empirical proxies. The theoretical object of Proposition 1 combines net dollar operating flow with liability exposure:

$$\mathcal{X} = \frac{\text{USD export revenue} - \text{USD input costs}}{\text{EBITDA}} - \text{USD debt share},$$

where dollar revenues are beneficial exposure and dollar inputs and dollar debt are adverse exposure. The data above measure the debt-share component directly and proxy the operating-flow component with industry tradability; they therefore establish that the currency composition of debt differs by tradability, and are consistent with — but do not directly measure — net short-dollar cash-flow positions in the non-tradable sector.

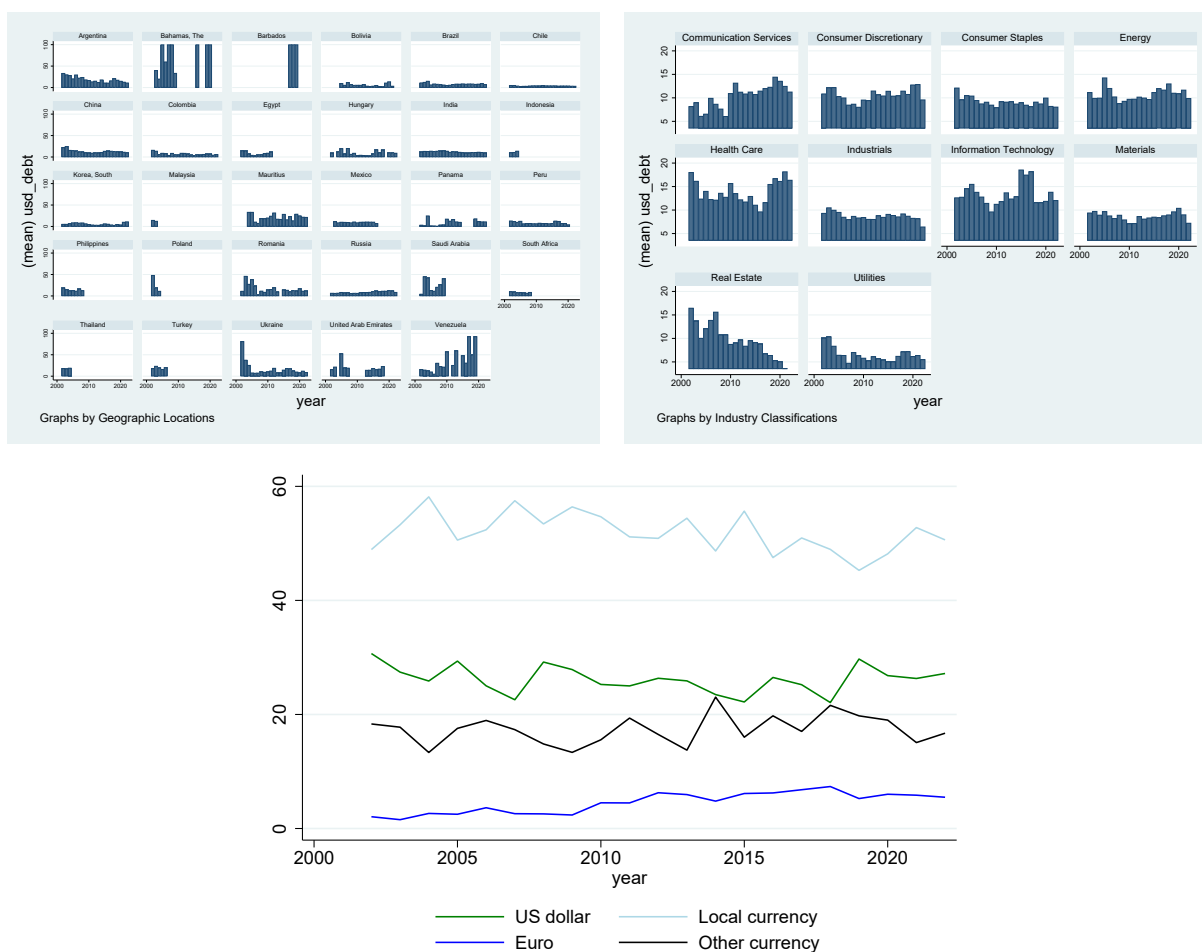


Figure 17: Capital IQ: dollar debt shares by country-year (top left), industry-year (top right), and over time (bottom).